

Rapid #: -24137480

CROSS REF ID: **638797**

LENDER: **GZO (University of Wisconsin, Oshkosh) :: Polk Library**

BORROWER: **VOD (Old Dominion University) :: Main Library**

TYPE: Article CC:CCG

JOURNAL TITLE: Nature methods

USER JOURNAL TITLE: Nature Methods

ARTICLE TITLE: DiffModeler: large macromolecular structure modeling for cryo-EM maps using a diffusion model

ARTICLE AUTHOR: Xiao Wang, Han Zhu, Genki Terashi, Manav Taluja &

VOLUME: 21

ISSUE: 12

MONTH: Dec

YEAR: 2024

PAGES: 2307-2317

ISSN: 1548-7091

OCLC #: 54447242

Processed by RapidX: 3/4/2025 10:13:01 AM

This material may be protected by copyright law (Title 17 U.S. Code)

DiffModeler: large macromolecular structure modeling for cryo-EM maps using a diffusion model

Received: 25 March 2024

Accepted: 19 September 2024

Published online: 21 October 2024



Xiao Wang¹, Han Zhu¹, Genki Terashi², Manav Taluja^{2,3} & Daisuke Kihara^{1,2}✉

Cryogenic electron microscopy (cryo-EM) has now been widely used for determining multichain protein complexes. However, modeling a large complex structure, such as those with more than ten chains, is challenging, particularly when the map resolution decreases. Here we present DiffModeler, a fully automated method for modeling large protein complex structures. DiffModeler employs a diffusion model for backbone tracing and integrates AlphaFold2-predicted single-chain structures for structure fitting. DiffModeler showed an average template modeling score of 0.88 and 0.91 for two datasets of cryo-EM maps of 0–5 Å resolution and 0.92 for intermediate resolution maps (5–10 Å), substantially outperforming existing methodologies. Further benchmarking at low resolutions (10–20 Å) confirms its versatility, demonstrating plausible performance.

Proteins are fundamental molecules that carry out numerous functions in living organisms, including enzyme catalysis, cell signaling and transport of molecules. Cryogenic electron microscopy (cryo-EM) has gained notable popularity among experimental protein structure determination techniques^{1–3}. This technique is increasingly favored owing to several advantages, notably its superior capacity to determine the three-dimensional structures of large macromolecular complexes.

While reported map resolutions in literature have generally shown steady improvement over recent years, it remains common to encounter intermediate resolutions (~5–10 Å) in real-life laboratory scenarios, posing challenges for structure modeling. When the map resolution is better than 5 Å, direct tracing of the main chain of proteins^{4–8} and nucleic acids⁹ have now become feasible due to recent modeling methods leveraging deep learning to detect atom positions within the map. However, for maps within the intermediate resolution range (5–10 Å), de novo modeling is generally not viable because the identification of amino acid residues and atoms remains elusive, even with deep learning techniques. Hence, a practical approach involves conducting structure fitting using methods such as Phenix¹⁰, Flex-EM¹¹, Assemblin¹², MultiFit¹³, Chimera¹⁴, MarkovFit¹⁵ and vector-based local space electron density map alignment (VESPER)¹⁶ or employing manual fitting with

known structures from the Protein Data Bank (PDB)¹⁷ or predicted structure models¹⁸. Secondary structure detection methods within electron microscopy (EM) maps^{19–21} can also aid in protein structure fitting. Despite many structures are determined through structure fitting, accurately orienting molecules within a map in this resolution range remains challenging, especially for complexes comprising multiple subunits. The successful development of an automatic and precise structure fitting method for EM maps at medium and low intermediate resolutions would substantially support structural biologists.

Here, we developed DiffModeler, a fully automated structure fitting method for modeling large protein complex structures in cryo-EM maps with resolutions up to about 15 Å. DiffModeler uses a diffusion model^{22–24} to enhance the map aiding in finding precise fitting poses for these structures. The diffusion model is a parameterized Markov chain trained using variational inference to generate samples that match the underlined data after a finite time frame. Notably, the diffusion model has demonstrated considerable success in various areas of image processing, such as image generation^{23–26}, segmentation^{27,28} and translation^{29,30}, as well as in bioinformatics, including protein docking³¹ and protein design^{32,33}. Building upon these successful applications, DiffModeler integrates the diffusion model to enhance the extraction

¹Department of Computer Science, Purdue University, West Lafayette, IN, USA. ²Department of Biological Sciences, Purdue University, West Lafayette, IN, USA. ³School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India. ✉e-mail: dkihara@purdue.edu

of structural information, facilitating accurate structure modeling for cryo-EM maps at intermediate resolutions.

To the best of our knowledge, this is the first fully automated and accurate method for modeling protein complex structures in maps at intermediate resolutions. DiffModeler initiates the process by tracing protein backbones within a cryo-EM map, employing a diffusion model designed to capture the distinctive local density patterns representing protein backbones. Simultaneously, we use AlphaFold2 (AF2)³⁴, the cutting-edge protein structure prediction method, to generate high-quality single-chain structures. Subsequently, the structure models from AF2 are fitted into the traced backbone map, producing many candidate poses through the VESPER¹⁶ structure fitting program. Ultimately, the complete protein complex structure is assembled by combining candidate poses of constituent subunits.

A benchmark conducted on EM maps ranging from 5.0 Å to 10.0 Å resolution demonstrated that modeling with DiffModeler substantially outperformed conventional methods^{10,16}. Extending our evaluation, we further benchmarked DiffModeler on six experimental maps at a low resolution of 10 to 20 Å, where DiffModeler modeled the structure with a template modeling (TM) score of 0.27–0.97. Additionally, we integrated DiffModeler with CryoREAD, our DNA/RNA structure modeling method⁹, to build protein–nucleic acid complex structures in two datasets comprising 61 and 28 maps at a resolution of up to 5 Å. This combined protocol showcased a state-of-the-art performance delivering an average TM score³⁵ of 0.88 and 0.91, respectively.

Results

Overall framework of DiffModeler

We begin by explaining the DiffModeler algorithm depicted in Fig. 1. DiffModeler comprises four major steps. First, it detects the protein backbone positions in the input cryo-EM map by enhancing the map using a trained diffusion model. Second, it conducts the modeling of individual protein structures using AF2. Third, structure models are fitted to the enhanced map using VESPER. Last, it selects and combines fitted single-chain poses to build the complete protein complex structures within the map. Below, we provide more information of each step.

Backbone tracing via the diffusion model. Achieving accurate structure fitting for maps of an intermediate resolution is untrivial. To aim for higher accuracy, the main innovation of DiffModeler is to use a diffusion model to pronounce the density that belong to protein backbone. The input map is scanned with a 64^3 Å³ box with a stride of 32 Å. Given a box of cryo-EM density, the encoder of the conditional diffusion model computes an embedding of the input density box. Subsequently, the decoder starts with random Gaussian noise as the initial density distribution and iteratively refines its estimates to make it closer to the ground-truth traced backbone conditioned on the embedding from the encoder and the initial density input. The diffusion process is illustrated in Fig. 1b. This traced backbone provides clearer information for structure fitting compared with the original map. The diffusion model is trained via denoising diffusion implicit model framework, with the main objective to perform conditional denoising of a noisy density of the traced backbone to achieve the ground-truth traced protein backbone density in the map. The overall framework is optimized via Dice loss³⁶ that considers the agreement of the identified and ground-truth backbone positions. The training and inference framework is presented in Extended Data Figs. 1–4, respectively, and further details can be found in Methods.

Structure prediction by AF2. In DiffModeler, we use predicted single-chain protein structures by AF2 (ref. 34) to fit into the diffused map. While there are instances where AF2 models do not align with the proteins' conformations in particular cryo-EM maps⁸, ample cases exist^{37–39} where AF2 models demonstrated sufficient accuracy to be effectively integrated into EM maps. Specifically, for maps at a

resolution worse than 5 Å, where de novo main-chain tracing becomes highly challenging, it would be pragmatic to consider AF2 models for structure modeling. Instead of generating new AF2 models, users can also use precomputed models available in the AlphaFold database¹⁸, which we employed in this work.

Structure model fitting with VESPER. The predicted structure models are fit to the diffused map using VESPER¹⁶, a structure and map fitting method developed in our group. By taking into account local density gradient within maps, VESPER has demonstrated superior performance, surpassing existing methods¹⁶. The predicted structure models are converted into simulated maps at a 1 Å resolution. Subsequently, both these simulated maps of derived from the models and the diffused backbone map are transformed into local dense points (LDPs) using the mean-shift algorithm^{4,40}. LDPs serve to encapsulate the local salient features of density, proving to be more precise for alignment than using the unprocessed maps. Using VESPER, each subunit is aligned with the diffused map and the top 100 candidate poses are kept for the subsequent assembly phase.

Protein complex modeling by a greedy assembling algorithm. This phase is geared toward assembling the complete protein complex structure through an assembly of suitable poses from each subunit. To accomplish this, we have devised a greedy algorithm, which is explained in detail in Methods and visually outlined in Extended Data Fig. 5. In the preceding step, a collection of 100 poses has been constructed for each subunit, with each pose being evaluated based on a fitness score. From all combinations of subunit–pose pairs, we identify the subunit–pose with the highest score. Subsequently, we mask the local density of the map occupied by this selected subunit–pose pair and select the next best subunit–pose in the pool. This process iterates, systematically selecting the subsequent best subunit–pose pairs until all subunits seamlessly integrate into the diffused protein backbone map.

Fitting quality estimation in DiffModeler. The quality of the structure fit is quantified by subunit_fitscore (equation (9) in Methods), used to rank fitting poses of chains. This score ranges from 0 to 1, with higher scores indicating better fitting quality. As illustrated in Extended Data Fig. 6 (EMD-21136, PDB 6vac), well-fitted structures typically have a score higher than 0.5, whereas chains with scores below 0.5 (depicted in red) indicate potentially incorrect fit.

Structure modeling performance at intermediate resolution

We assessed DiffModeler's performance on an independent dataset comprising 71 maps determined at resolutions between 5.0 Å and 10.0 Å. These maps included 19 cases where all proteins in the map were predicted by AF2 with a TM score no worse than 0.5, whereas the other 52 maps include inaccurate AF2 models. These structures are nonredundant in comparison with the training and validation datasets we used (Methods). Supplementary Table 1 provides a comprehensive list of the maps included in this dataset. The range of residues of proteins in the 19 maps varied from 1,202 to 13,462, while the number of protein chains ranged between 3 and 47. Notably, 12 out of 19 maps include protein complexes with more than 3,000 residues in total, which is larger than the size that the state-of-the-art protein docking method, Alphafold-Multimer⁴¹ was trained on. The number of chains in the other 52 maps ranged from 1 to 62, with the number of residues ranging from 814 to 18,080.

Figure 2 summarizes the modeling accuracy on the dataset from various perspectives. In Fig. 2a, we assessed the accuracy of the diffused backbone map generated by the diffusion model in the initial step of DiffModeler (as depicted in the traced backbone image in Fig. 1). We computed recall and precision of the grid points within the diffused map with reference to the backbone heavy atoms (C α , C and N) excluding oxygen of proteins in a map (details in Methods). As a diffused map

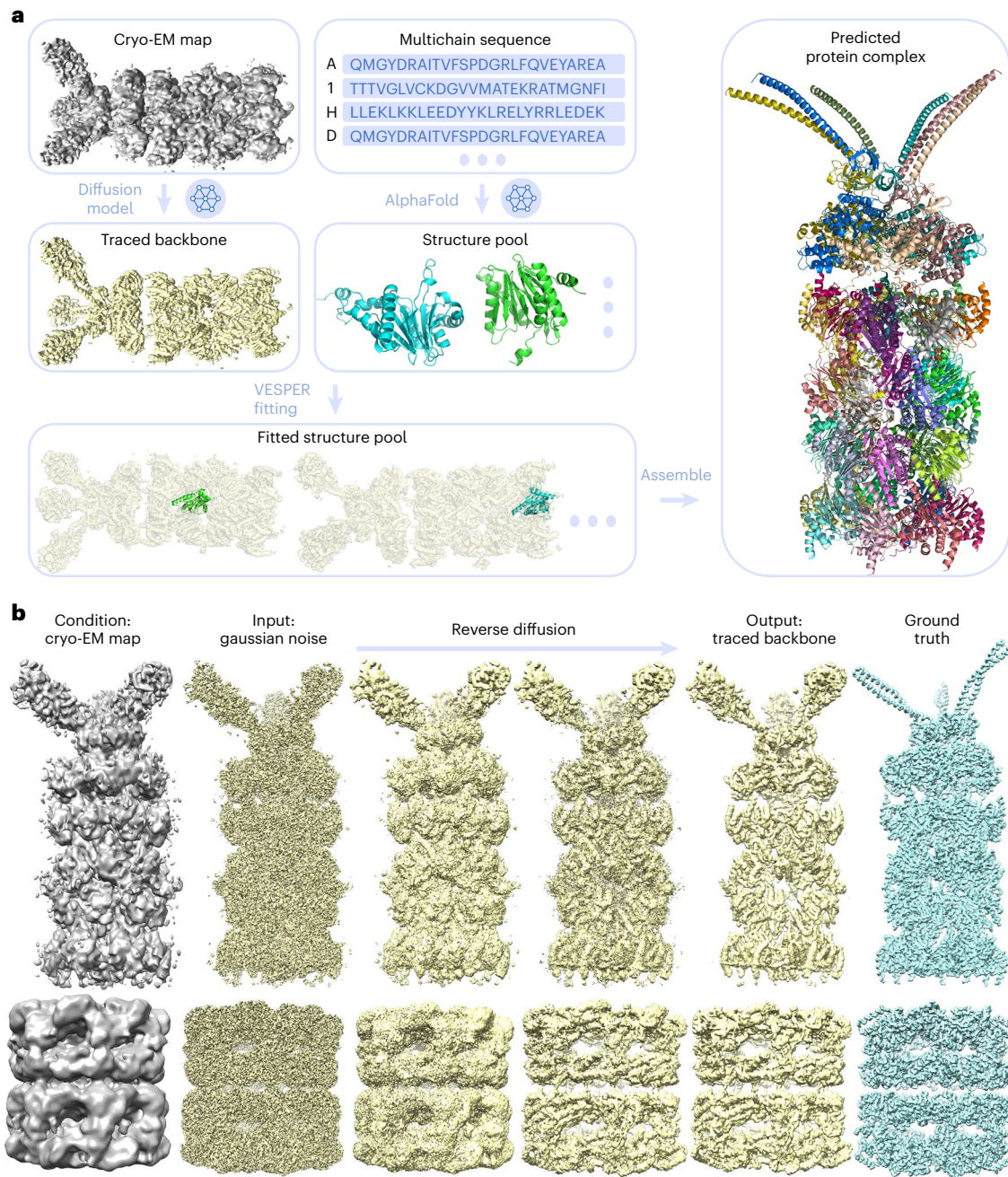


Fig. 1 | Overall framework of DiffModeler. a, The workflow of DiffModeler. DiffModeler consists of four main steps: (1) backbone tracing from cryo-EM maps at intermediate resolution via a diffusion model, (2) single-chain structure prediction by AF2 (example input chains: A, I, H, D), (3) single-chain structure fitting using VESPER and (4) protein complex modeling by assembling algorithms. **b**, An overview of the diffusion process. Starting from an original

cryo-EM map as a condition and random Gaussian noise, the protein backbone is traced by the iterative reverse diffusion process utilizing a pretrained diffusion model. On the right is the ground-truth protein backbone density, which is the target of the reverse diffusion process. The examples used here are EMD-0213 (resolution 6.35 Å) and EMD-1042 (resolution 10.3 Å).

outlines backbone atom positions within an input EM map, the volume of a map was, in principle, reduced, on average, by 53.7% for the 19 maps (which did not have inaccurate AF2 models). This modification of maps notably elevated the average precision to 85.1% from 68.8% without compromising the recall, which remained stable at an average of 93.1% from 96.6% (the original maps). Detailed results of individual maps are available in Supplementary Table 2. Precision and recall for the 52 maps with inaccurate AF2 models (shown as orange crosses) were at similar levels, of 80.6% and 93.5%, respectively.

Figure 2b illustrates our exploration into the impact of backbone recall within diffused maps on the subsequent accuracy of structure

fitting. To assess the precision of modeled protein complexes, we employed multimer (MM)-align⁴² to superimpose a modeled complex structure onto the accurate structure (referencing the PDB entry associated with the map) and calculated the TM score³⁵ (details in Methods). The TM score is a dimensionless metric utilized to gauge structural resemblance between two protein structures, with a value of 1 denoting identical protein pairs and values exceeding 0.5 indicative of meaningful similarity. When the 19 maps with no incorrect (TM score < 0.5) AF2 models were considered (blue circles in the plot), on average, DiffModeler achieved a high TM score of 0.808. There were two instances where the TM score fell below 0.8. In a particular

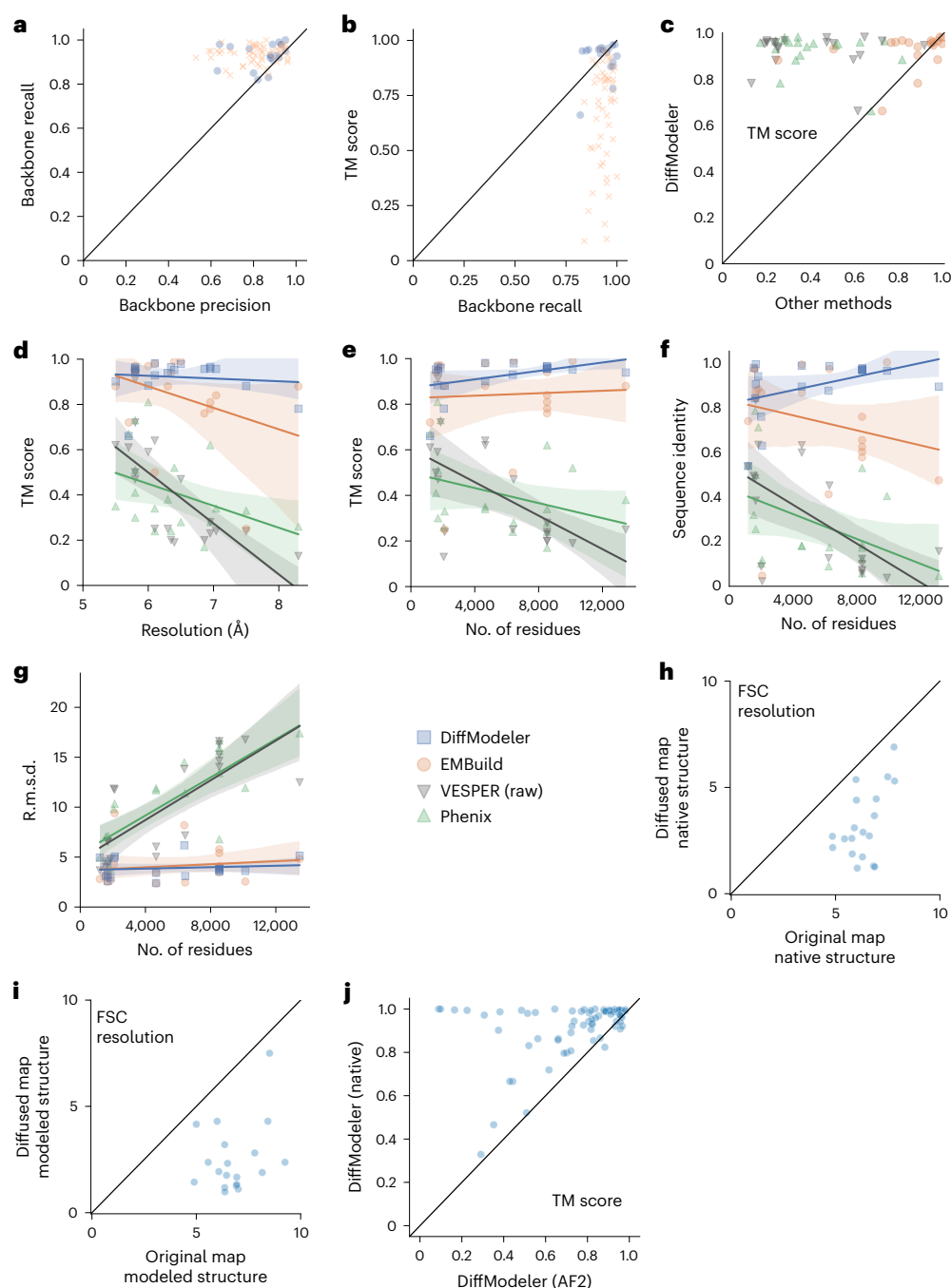


Fig. 2 | Performance of protein complex structure modeling by DiffModeler.

a, Backbone recall and precision of the diffusion model. Recall and precision were computed by considering grid points in the diffused maps relative to ground-truth positions of main-chain atoms in the maps. Details in Supplementary Table 2. Nineteen maps with no inaccurate AF2 models (TM score < 0.5) are shown with blue circles, while 52 maps that include inaccurate AF2 models are shown with orange crosses. **b**, The TM score of the modeled protein complex structure relative to backbone recall. The symbols are the same as in **a**. **c**, TM score comparison between DiffModeler and three existing methods: the raw VESPER, Phenix and EMBuild. **d**, TM score relative to the map resolution of the different methods. The lines represent the regression line, and the shaded region represents the confidence interval for the regression estimate. **e**, TM

score relative to the overall protein complex size represented by the number of residues for the different methods. **f**, The sequence identity relative to the structure size for the different methods. **g**, The r.m.s.d. relative to the structure size for the different methods. **h**, FSC resolution estimation between diffused maps and original maps relative to the native structure in PDB. The resolution was estimated at FSC of 0.143. **i**, FSC resolution estimation comparison between diffused maps and original maps relative to the modeled structure by DiffModeler. Raw data of the plots are available in Supplementary Table 3. **j**, Modeling results by DiffModeler using the native single-chain structures as compared with the results using AF2 models. Here, 71 maps were used. For statistical information of regression lines in **d–g**, see Methods.

case (EMD-1871), despite a high backbone recall of 0.98 (close to 1.0), the TM score remained at 0.781 (close to 0.8), owing to two wrong single-chain structure fittings because of the low backbone precision 0.64.

We further evaluated DiffModeler on all 71 test maps, which included 52 maps with incorrect AF2 models (TM score < 0.5) (crosses in the plot). The average TM scores of modeled structures decreased from 0.922 to 0.737. However, 59 out of the 71 maps (83.1%) still maintained

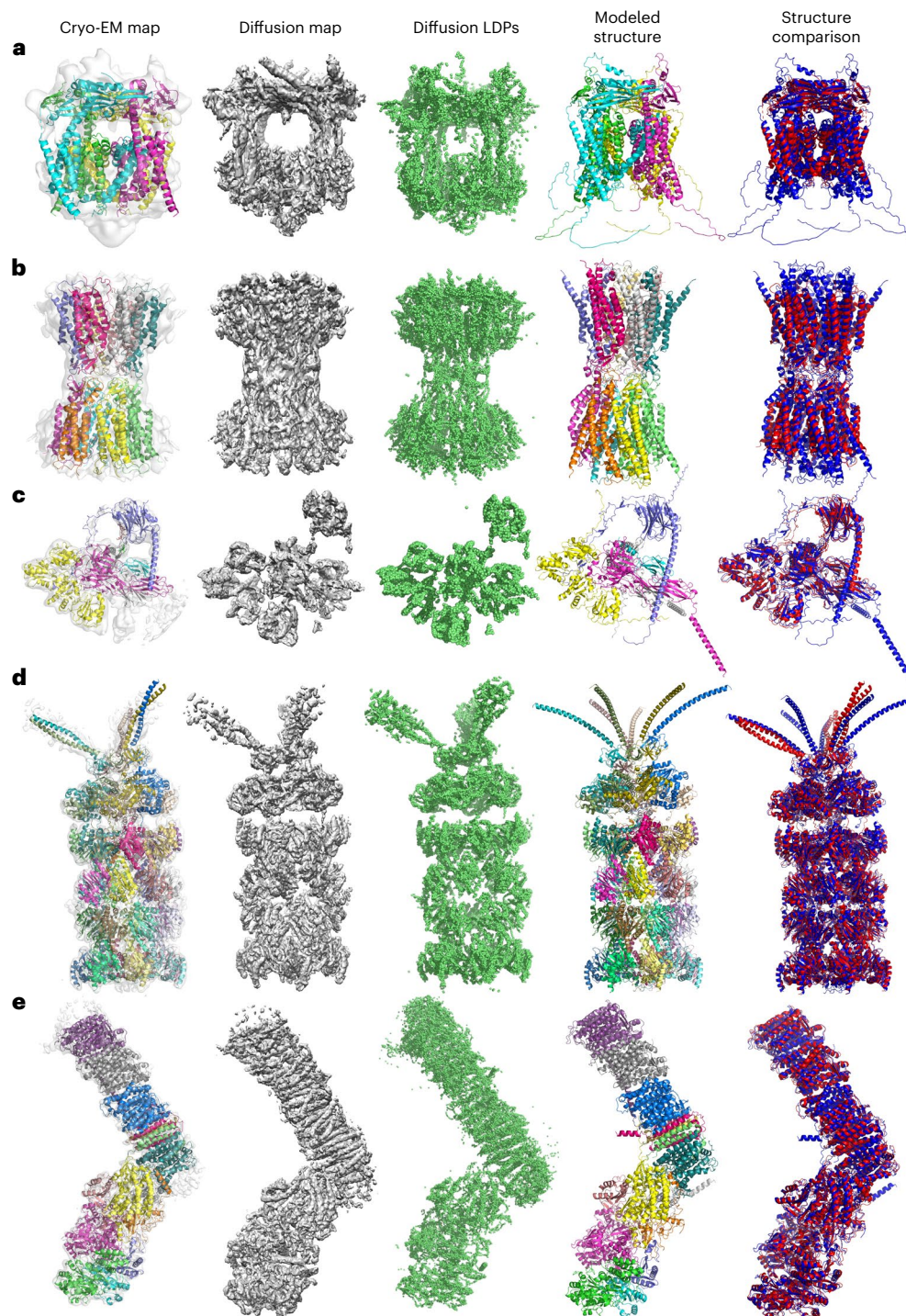


Fig. 3 | Examples of structure models constructed by DiffModeler from the test dataset. Detailed evaluation results are presented in Supplementary Table 3. For each example, five columns are shown: the cryo-EM map with the structure of the protein complex with different color indicating different chains; the diffused backbone map by the diffusion model; the local dense points of the backbone map; the structure model by DiffModeler and the superposition of the model by DiffModeler (blue) with the native structure (red). **a**, State 2 of *M. musculus* TRPML1 (EMD-6824, PDB 5YE1; resolution 7.4 Å (see text); protein size 4 chains and 1,696 amino acids (aa)). TM score 0.95; r.m.s.d. 3.26 Å. **b**, The

closed conformation of Cx26 Gap junction channels at acidic pH (EMD-20916, PDB 6UVT; resolution 7.50 Å; protein size 12 chains and 2,112 aa). TM score 0.88; r.m.s.d. 5.04 Å. **c**, The human peptide-loading complex editing module (EMD-3906, PDB 6ENY; resolution 5.80 Å; protein size 5 chains and 1,569 aa). TM score 0.95; r.m.s.d. 3.08 Å. **d**, State 2 of ATPase cycle in proteasome-activating nucleotidase (EMD-213, PDB 6HE9; resolution 6.35 Å; protein size 34 chains and 8,531 aa). TM score 0.97; r.m.s.d. 3.79 Å. **e**, The minor state of *T. thermophilus* enzyme in complex with NADH (EMD-11237, PDB 6ZJN; resolution 6.10 Å; protein size 15 chains and 4,655 aa). TM score 0.98; r.m.s.d. 2.40 Å.

TM scores higher than 0.5, indicating that the modeled structures shared similar folds with the native structures. Among the 52 maps, there were 3 instances where the overall TM score fell below 0.2. In all these cases, multiple AF2 models in a map were incorrectly built. For

example, in the case of EMD-9036, the map with the smallest TM score of 0.056 includes only one chain in the map that had a TM score of 0.089.

In Fig. 2c, we compared the TM score of models constructed by DiffModeler with three other existing methods: the dock_in_map

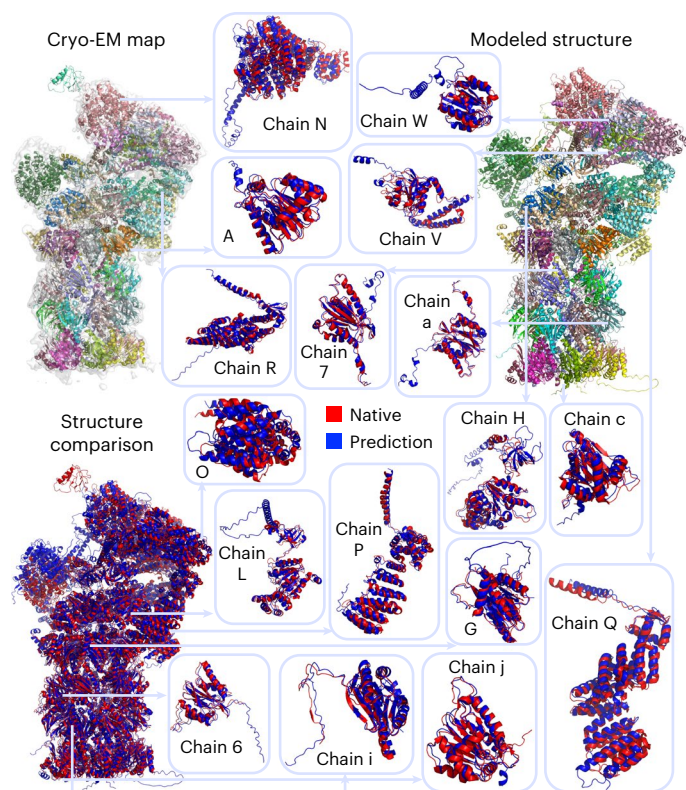


Fig. 4 | Structure model of proteasome constructed by DiffModeler. This is the largest complex in the test set of a proteasome in complex with ADP-AIFx (EMD-6693, PDB 5WVI; resolution 6.30 Å; 47 chains and 13,462 aa). TM score 0.94; sequence identity 0.89; r.m.s.d. 5.13 Å. The EM map superimposed with the corresponding complex structure and the model by DiffModeler are shown on the top left and top right, respectively, with different colors indicating different chains. On the bottom left, superimposition of the entire complex in PDB and the model is shown together with comparison of 17 individual chain models that appear in the front view of the complex. Blue is the model, and red is the native structure in the PDB entry. The modeled structures by the other methods are shown in Extended Data Fig. 7.

program in Phenix¹⁰, EMBuild⁴³ and raw VESPER¹⁶. For the latter, the original EM maps were used instead of the diffused maps for structure fitting. EMBuild is a recent method for fitting AF2 models within a cryo-EM map, which combines structure fitting, domain-based refinement and graph-based iterative assembly. DiffModeler exhibited a high average TM score of 0.922. In contrast, VESPER (raw), Phenix and EMBuild showcased a broad spectrum of model accuracy, averaging approximately half of DiffModeler's performance, with TM scores of 0.407, 0.409 and 0.841, respectively. The notable contrast between DiffModeler and VESPER (raw) vividly highlights the substantial positive impact of utilizing diffused maps.

Figure 2d–g aims to explore the relationship between model accuracy and both map resolution (Fig. 2d) and the size of the complexes (Fig. 2e–g). While the performance of other methods noticeably declined with increasing resolution and larger structure sizes, DiffModeler consistently maintained stable performance and notably outperformed in challenging scenarios involving lower resolutions or larger sizes. Figure 2f compares the sequence identity of different methods relative to the structure size, which considers the fraction of residues in the reference structure that were successfully modeled and with the correct residue type. DiffModeler demonstrated the stable sequence identity, while all other methods decreased dramatically when the structure size was large. On average, DiffModeler, VESPER (raw), Phenix and EMBuild yielded sequence identities of 0.89, 0.31,

0.29 and 0.74, respectively. Figure 2g investigates model accuracy concerning complex sizes using a different metric, the root mean standard deviation (r.m.s.d) of the aligned residues in the model (for details, see Methods). On average, DiffModeler, VESPER (raw), Phenix and EMBuild yielded r.m.s.d. values of 3.89 Å, 10.09 Å, 10.48 Å and 4.08 Å, respectively (details in Supplementary Table 3). EMBuild showed lower sequence identity (Fig. 2f) in general than DiffModeler, but their r.m.s.d. values are comparable (Fig. 2g). This indicates that EMBuild places chains in structurally similar regions, yielding a small r.m.s.d, which are, however, not the correct native positions, which resulted in lower sequence identity.

In Fig. 2h,i, we examined map–model agreement, calculating the Fourier shell coefficient (FSC) with phenix.validation_cryoem⁴⁴. Figure 2h examines the original experimental maps and the diffused maps with the native structure, while Fig. 2i compares the original and diffused maps with the modeled structures by DiffModeler. For both cases, a clear improvement was observed when comparing the diffused map with the structures.

In Fig. 2j, we show the modeling results by DiffModeler using the native single-chain structures in comparison with results using the AF2 models because the inaccuracy of AF2 models is a main reason for low accuracy of DiffModeler models. As shown in the plot, most of the 71 testing maps have a high TM score by using the native chain structures. The average TM score improved from 0.737 to 0.917.

Examples of protein complex structure models

In this section we discuss five examples of models constructed by DiffModeler. In Fig. 3, for each example map, five images are shown: the original experimental map, the diffused backbone map, LDPs of traced backbone, the structure models and structure comparison between the constructed model with the PDB entry. The first example (Fig. 3a) is the state 2 of *Mus musculus* TRPML1 (EMD-6824; resolution 7.4 Å), which encompasses four protein chains totaling 1,696 residues⁴⁵. The resolution of this map was mentioned to be 7.4 Å in the paper⁴⁵ but it may be even worse because Resmap⁴⁶, a map resolution estimation program, reported 9.4 Å when we ran it. Modeling the interaction between the transmembrane domain and the peripheral domain was particularly difficult for this map, resulting in low TM scores of 0.30 and 0.47 using Phenix and VESPER (raw), respectively. In contrast, DiffModeler nicely traced the backbone by diffusion model, achieving a TM score of 0.95 and an align ratio of 1.0 for this challenging map.

The next example (Fig. 3b) is the closed conformation of Cx26 Gap junction channels at acidic pH (EMD-20916)⁴⁷. This complex is difficult to model because it has 12 chains in a map of a relatively low resolution (7.5 Å). DiffModeler was able to precisely identify helices in the map and correctly fit the 12 chains with a TM score of 0.88. In contrast, VESPER (raw) struggled to find correct poses of the chains, resulting in a TM score of 0.24.

Figure 3c is the model for the human peptide-loading complex editing module (EMD-3906; resolution 5.8 Å)⁴⁸. Modeling the full protein complex is difficult due to the substantial flexibility exhibited by calreticulin (the chain in purple) and the sparseness of the chain assembly. Fitting the structures to the original experimental map was challenging, as indicated by a low TM score of 0.5 by VESPER (raw). In contrast, DiffModeler achieved a high TM score of 0.95, demonstrating that the diffusion model was effective to capture structural features in the map.

The next map is a complex with 34 chains (Fig. 3d). It is the state 2 of a complex of the proteolytic core and the ATPase proteasome-activating nucleotidase (EMD-213; resolution 6.35 Å)⁴⁹. DiffModeler was able to fit most of the subunits correct except for long helical domains locating at the top of the complex in the figure, yielding a TM score of 0.97. In comparison, with the original map, VESPER (raw) was only able to fill about 20% of the structure, with a TM score of 0.20.

The last example (Fig. 3e) is the minor state of the *Thermus thermophilus* enzyme in complex with NADH (EMD-11237; resolution 6.10 Å)⁵⁰,

which includes 15 chains. Fitting subunit structures to the original map was difficult as all the chains are α -helical and hard to distinguish, as indicated by a low TM score of 0.64 by VESPER (raw). On the other hand, with the advantage of the map diffusion, DiffModeler showed accurate backbone structure tracing with backbone recall of 0.98 and a superior structure alignment with a TM score of 0.98 and an r.m.s.d. of 2.40 Å.

Figure 4 illustrates the largest protein complex structure built by DiffModeler. This example is proteasome in complex with ADP-AIFx (EMD-6693; resolution 6.30 Å)⁵¹. The complex comprises 47 protein chains totaling 13,462 amino acids. The diffusion model in DiffModeler achieved a 0.92 backbone tracing recall, laying a robust foundation for further protein complex structure modeling. Overall, the modeled complex showed high consistency with the native structure, as evidenced by a TM score of 0.94 and a sequence identity of 0.89. When individual chains are considered, 45 chains out of 47 chains were successfully modeled, with an average sequence matching of 92.6%. Seventeen out of 45 individual chain structures are shown in the figure, which appear in the front view of the complex. The high modeling accuracy is clearly due to the application of the diffusion model to the map, as VESPER (raw) alone only achieved a TM score of 0.25. In contrast, EMBuild yielded a TM score of 0.88 and a sequence identity of only 0.47, which indicates many chains were placed on similar but incorrect map regions.

Structure modeling on cryo-EM maps at low resolution

We further conducted an additional benchmark of DiffModeler on cryo-EM maps determined at low resolutions (10–18 Å). There were four maps in the Electron Microscopy Data Bank (EMDB) in this resolution range and that satisfy the map selection criteria we used (Methods). The modeling results are shown in Fig. 5 and detailed performance metrics are provided in Supplementary Table 4. For these four maps, the average TM score of models by DiffModeler was 0.74, while those of EMBuild, Phenix and VESPER (raw) were 0.32, 0.36 and 0.27, respectively, which are below the cutoff of 0.5 that indicates meaningful structural similarity.

The first example (Fig. 5a) is the ATP-bound states of GroEL (EMD-1042; resolution 10.3 Å; 14 chains, 7,238 residues)⁵². Owing to the low resolution of the map, the authors manually determined this structure by fitting individual chain structures while considering symmetry information. In contrast, DiffModeler demonstrated the capability to model the complete atomic structure automatically and accurately, achieving a TM score of 0.97 and an r.m.s.d. of 3.88 Å. The structural superimposition of the model with the corresponding PDB entry visually confirms the accuracy of the model.

Figure 5b is a 10.3 Å map from anaerobic fatty acid beta oxidation trifunctional enzyme (anEcTFE) octameric complex (EMD-16134)⁵³. The complex has eight chains, which is a dimer of a tetramer, shown as left and right volumes in the map in the figure. The original investigators modeled the complex structure with multiple manual steps. The procedure included structure fitting from a related tetramer map with a resolution of 3.55 Å, which was modeled by incorporating the crystal structure (PDB 6DV2 (ref. 54)) with further fitting and refinement. Subsequently, they docked the solved structure into the low-resolution map and conducted further refinement to achieve the final structure. In contrast, DiffModeler automated the assembly of the entire protein complex based on the low-resolution map and achieved a high TM score of 0.87. Structures derived from EMBuild, Phenix and VESPER reported TM scores of 0.30, 0.30 and 0.19, respectively, emphasizing the distinct advantage of DiffModeler.

The third map was cofilactin filament inside microtubule lumen (EMD-16877), determined even at a lower resolution of 16.5 Å with 14 chains (Fig. 5c). The authors determined the structure by fitting cofilactin filament model (PDB 5YU8) to the density manually followed by a local refinement⁵⁵. The model built by DiffModeler had a TM score of 0.60, which was substantially higher than values of EMBuild, Phenix

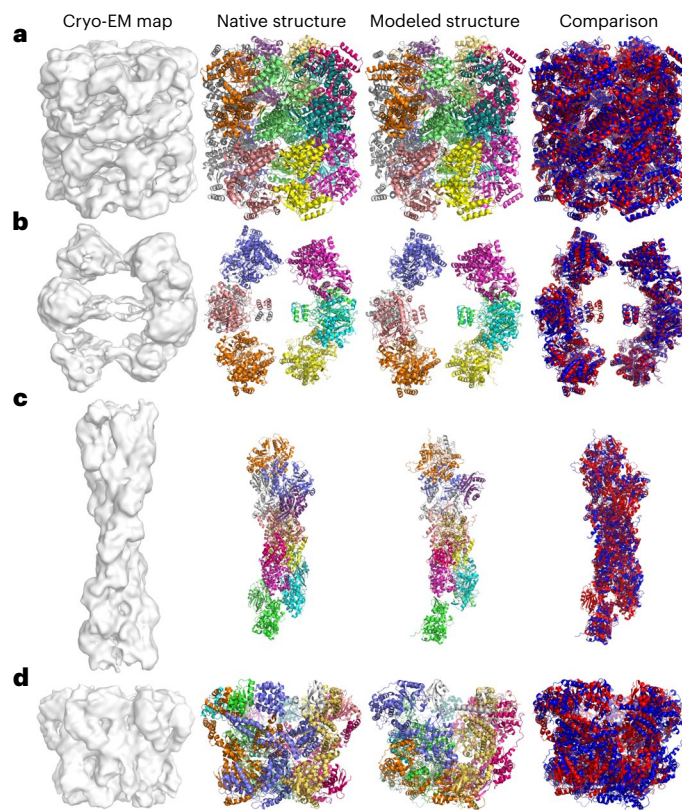


Fig. 5 | Modeling results by DiffModeler for experimental maps at low resolution. Detailed evaluation results are presented in Supplementary Table 4, which contains the TM scores of individual chains and modeled protein complexes by different methods. For each map, four images are shown, from left to right: the input cryo-EM map, the corresponding native structure, the model by DiffModeler and the superposition of the DiffModeler model (blue) with the native structure (red). **a**, ATP-bound states of GroEL (EMD-1042, PDB 1GR5; resolution 10.3 Å; 14 chains and 7,238 aa). TM score 0.97; sequence identity 0.95; r.m.s.d. 3.88 Å. **b**, Anaerobic fatty acid beta oxidation trifunctional enzyme (anEcTFE) octameric complex (EMD-16134, PDB 8BNN; resolution 10.3 Å; 8 chains and 4,584 aa). TM score 0.87; sequence identity 0.88; r.m.s.d. 5.80 Å. **c**, Cofilactin filament inside microtubule lumen (EMD-16877, PDB 8OH4; resolution 16.5 Å; 14 chains and 3,776 aa). TM score 0.60; sequence identity 0.50; r.m.s.d. 8.30 Å. **d**, Meca-ClpC complex with ATP with the Walker B mutations introduced in the D2 ring (EMD-5608, PDB 3J3S; resolution 11.0 Å; 12 chains and 5,352 aa). TM score 0.51; sequence identity 0.45; r.m.s.d. 8.42 Å. Modeling results by other methods are provided in Extended Data Fig. 8.

and VESPER (raw), at 0.17, 0.25 and 0.18, respectively, which failed to capture even the overall fold. The model by DiffModeler captured the overall shape of the complex. However, only 4 chains out of 14 chains are successfully aligned (sequence identity 0.99). There were chains, for example, chains E, H and K, which were placed in the correct regions but with incorrect alignments.

In Fig. 5d, we illustrate a case where DiffModeler's performance was relatively poor. The presented structure is derived from a 11.0 Å resolution map or a 12-chain complex of Meca-ClpC with ATP and Walker B mutations introduced in the D2 ring (EMD-5608). The authors employed a complex manual procedure for structure determination. Initially, they used an initial model based on another crystal structure of ClpC (PDB 3PXi) and employed MODELLER⁵⁶ to fill in missing loops using other related structures as templates (PDB 1JBK and 1R6B). Subsequently, the structure was manually docked into the cryo-EM maps, followed by flexible fitting using Nanoscale Molecular Dynamics (NAMD)⁵⁷. The model generated by DiffModeler had an overall TM score of 0.51, a barely significant score for structure modeling. Among

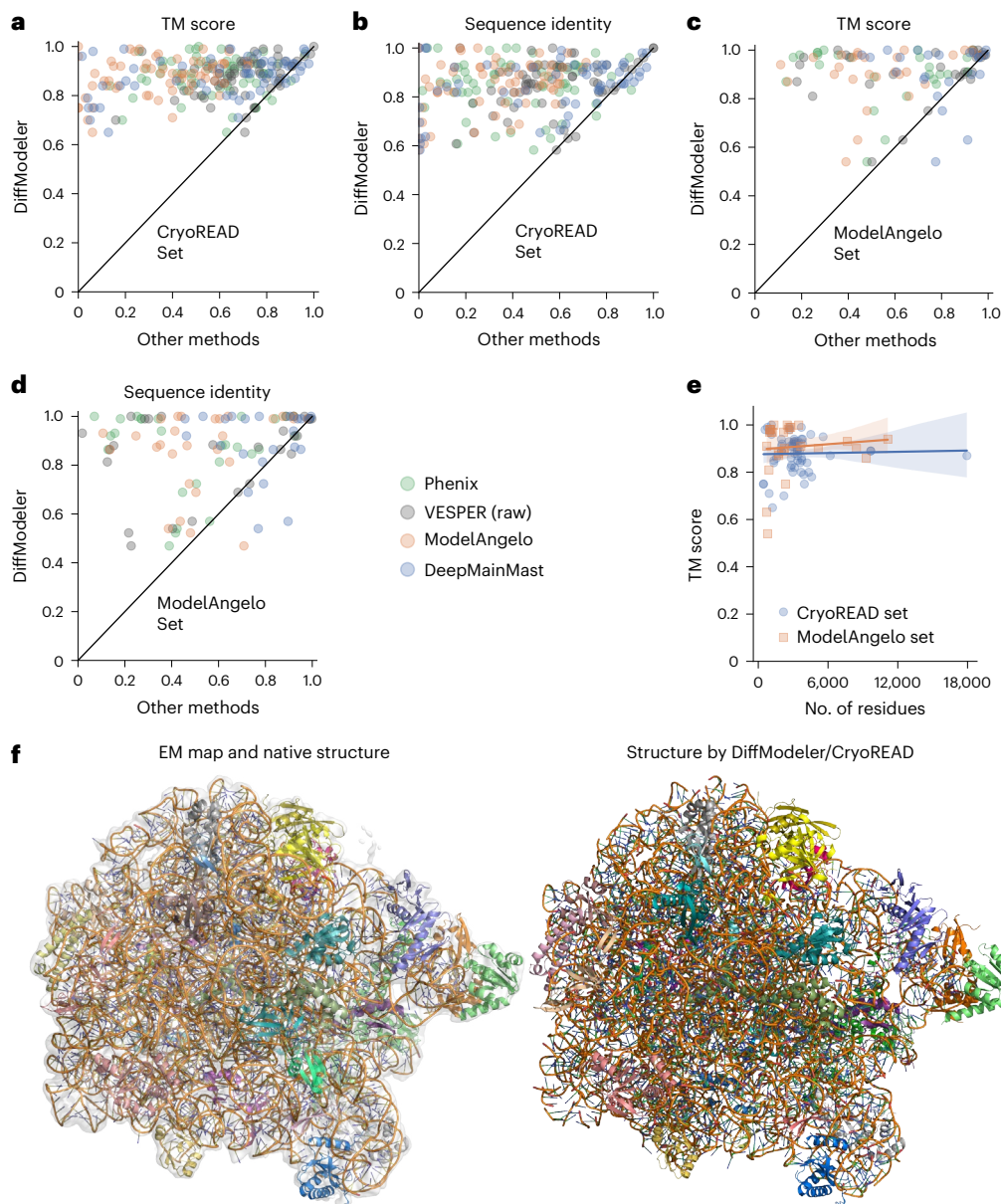


Fig. 6 | Protein complex structure modeling by DiffModeler for experimental maps at near-atomic resolution (<5 Å). Detailed evaluation results are presented in Supplementary Tables 5 and 6. The benchmark was performed on two datasets: the CryoREAD dataset and the ModelAngelo dataset. Models by DiffModeler were compared with those by Phenix, VESPER (raw), ModelAngelo and DeepMainMast. **a**, TM score comparison on the CryoREAD dataset. **b**, The sequence identity comparison on the CryoREAD dataset. **c**, TM score comparison on the ModelAngelo dataset. **d**, The sequence identity comparison on the ModelAngelo dataset. **e**, TM score relative to the total number of

residues in the map. For the CryoREAD dataset, the equation of regression line is $y = 8.70 \times 10^{-7}x + 0.876$ (Pearson correlation coefficient 0.029; P value 0.824; s.e.m. 3.90×10^{-6}). For the ModelAngelo dataset, the equation of regression line is $y = 3.71 \times 10^{-6}x + 0.896$ (Pearson correlation coefficient 0.095; P value 0.632; s.e.m. 7.67×10^{-6}). **f**, RqcH DR variant bound to 50S-peptidyl-tRNA-RqcP RQC complex (EMD-13017, PDB 70PE; resolution 3.2 Å; 3,818 residues and 2,996 nucleotides). DiffModeler and CryoREAD: TM score 0.92; sequence identity (proteins) 0.92; r.m.s.d. 1.74 Å.

12 chains, 4 chains C, D, E and F were modeled successfully, with an average TM score of 0.73 and sequence identity of 0.73. The rest of the chains were placed to incorrect regions of the map. TM scores of EMBuild, Phenix and VESPER (raw) were even worse, at 0.26, 0.30 and 0.19, respectively.

To our knowledge, DiffModeler is the first method capable of automatically modeling protein complexes from maps in this low-resolution range. It distinctly demonstrates its advantage over existing methods.

Structure modeling for maps at high resolution

Although the primary focus of DiffModeler is low-resolution maps where it can demonstrate its unique strengths, it also performs

effectively with higher-resolution maps. To illustrate this versatility, we employed DiffModeler on maps with better than 5 Å resolution. We conducted benchmarking on two distinct datasets: one that was used in the paper of CryoREAD⁹ and the other employed in ModelAngelo⁵⁸. These two datasets cover a broad spectrum of structures, encompassing protein–DNA/RNA complexes and protein-only configurations. The CryoREAD dataset comprised 61 maps (excluding those DNA/RNA-only maps), while the ModelAngelo dataset included 28 maps. The number of chains in the dataset ranged from 1 to 48 chains, totaling residues from 447 to 17,947. On this dataset we used the identical model and pipeline of DiffModeler without any alterations. For maps with protein–DNA/RNA complexes, we first used CryoREAD to construct

DNA/RNA structures and then modeled protein structures in the remaining regions in the maps by DiffModeler. AF2 models used in the DiffModeler modeling was selected from the AF2 database by BLAST⁵⁹ sequence search. Supplementary Table 5 shows the sequence identity and the TM score of AF2 models relative to the native structure of individual chains in the maps. The TM score of AF2 models ranged from 0.134 to 0.998, with an average TM score of 0.858 and 0.896 for CryoREAD and ModelAngelo datasets, respectively.

Figure 6 summarizes the modeling results, with details in Supplementary Table 6. In Fig. 6a–d, models of the maps were evaluated with the TM score and the sequence identity with other methods on the two datasets. We compared other four modeling methods: Phenix (phenix.dock_in_map), VESPER (raw), ModelAngelo⁵⁸ and DeepMainMast⁸. Among these methods, DiffModeler clearly outperformed the other three methods on the two datasets. The average TM score and the sequence identity by DiffModeler for the CryoREAD/ModelAngelo datasets were 0.879/0.907 (Fig. 6a,c) and 0.851/0.864 (Fig. 6b,d), respectively, which are comparable results as benchmarked on the original dataset of 5.0–10.0 Å resolutions (Fig. 2). In contrast, the other four methods showed substantially lower performance: Phenix, VESPER (raw), ModelAngelo and DeepMainMast⁸ yielded the average TM score, and the sequence identities were 0.572/0.697/0.348/0.597 and 0.430/0.579/0.328/0.555 on the CryoREAD dataset, and 0.573/0.701/0.542/0.791 and 0.508/0.605/0.532/0.779 on the ModelAngelo dataset, respectively. In Fig. 6e, we investigated the impact of complex size on the TM score of DiffModeler models. As depicted in Fig. 2, we consistently observed high TM scores, even for large complexes.

While DiffModeler exhibited strong performance for most cases, there were instances where the performance is low, indicated by TM scores or sequence identity values lower than 0.6. One contributing factor to these cases was the failure in predicting the AF2 chain structure (for example, EMD-12935 and EMD-27705; Supplementary Table 6). The TM scores of these two maps were 0.65 and 0.54, respectively, but if we used the native chain structures as input, they both improve to 0.99 (Extended Data Fig. 9). There are also two cases with a low sequence identity and a high TM score (for example, EMD-13619 and EMD-13620), which are cases of hetero-oligomers where subunits were placed in equivalent places of different chains.

In Fig. 6f, we show a model for the RqcH DR variant bound to 50S-peptidyl-tRNA-RqcP ribosome-associated protein quality-control (RQC) complex (EMD-13017; resolution 3.2 Å)⁶⁰. This large complex includes 3,818 amino acid residues and 2,996 nucleotides. We modeled the entire complex with DiffModeler and CryoREAD⁹ for protein and RNA, respectively, which yielded a TM score of 0.92 for protein and a backbone recall of 0.94 for RNA. While this work primarily focuses on protein structure modeling with DiffModeler, we also extended our modeling efforts to include nucleic acid structures within these maps using CryoREAD⁹. Backbone and sequence recall (see the methods section in the CryoREAD paper⁹), were measured at 0.855 and 0.523 in the CryoREAD dataset and 0.829 and 0.413 in the ModelAngelo dataset.

Discussion

DiffModeler is a structure modeling method that uniquely targets cryo-EM maps with low resolution of 5–15 Å. Within this target resolution range, the presence of noisy density in cryo-EM maps makes it exceedingly difficult to detect precise atom and amino acid positions as well as main-chain conformations in the map. DiffModeler overcomes these obstacles by sculpting out main-chain conformations from low-resolution maps using a diffusion model, which enables to achieve substantially higher accuracy in structure fitting. The benchmark of DiffModeler at higher resolution, better than 5 Å, further indicates its generalizability and accuracy to handle maps with higher resolution.

For the training of DiffModeler, we opted to train our models using experimental low-resolution EM maps and then benchmarked it on high-resolution settings instead of using Gaussian noise-simulated

low-resolution maps. This is because we found Gaussian noise does not properly simulate experimental low-resolution maps^{19,20} (more details in Supplementary Notes).

Although DiffModeler has demonstrated overall accuracy and effectiveness, it is crucial to address the limitations of the current version. First, in some regions with low local resolution, the backbone tracing of diffusion model may be inaccurate, leading to incorrect structure fitting. To address this issue, further enhancements can be made to prioritize the fitting of regions with higher local resolution, mitigating the risk of such errors. Likewise, if local EM density is missing for an entire or a part of a subunit, such subunit is difficult to model because there is no density that the diffusion model can modify. Second, as of now, DiffModeler exclusively supports protein structure complex modeling. To expand its applicability, future developments will aim to extend its capabilities to support protein/DNA/RNA complex structure modeling, enhancing its versatility in addressing a wider range of biological systems. Furthermore, for high-resolution cryo-EM maps (better than 4 Å), it will be essential to develop local structure refinement approaches that leverage the density information to refine predicted structures, further enhancing accuracy and reliability. Addressing these limitations remain as future developments. Last, the results obtained from DiffModeler fitting are inevitably influenced by the (in)accuracy of AF2 models of proteins included in the maps, which can be overcome by fitting individual domains of proteins because AF2 tends to build accurate domain structures (implements and examples included in Methods). The overall accuracy of the models is expected to improve as protein structure prediction methods become more accurate in the near future.

We firmly believe that DiffModeler will prove to be an indispensable and user-friendly tool for protein complex structure modeling, bridging a crucial gap in the availability of tools suitable for maps at low resolutions. The approach will also be applicable for cryo-electron tomography within the same resolution range, better than 15 Å, which is now increasingly available^{61,62}.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41592-024-02479-0>.

References

- Bai, X.-C., McMullan, G. & Scheres, S. H. How cryo-EM is revolutionizing structural biology. *J. Mol. Biol.* **40**, 49–57 (2015).
- Wüthrich, K. The way to NMR structures of proteins. *Nat. Struct. Biol.* **8**, 923–925 (2001).
- Adams, P. D. et al. PHENIX: building new software for automated crystallographic structure determination. *Acta Crystallogr. D* **58**, 1948–1954 (2002).
- Terashi, G. & Kihara, D. De novo main-chain modeling for EM maps using MAINMAST. *Nat. Commun.* **9**, 1618 (2018).
- Pfaff, J., Phan, N. M. & Si, D. DeepTracer for fast de novo cryo-EM protein structure modeling and special studies on CoV-related complexes. *Proc. Natl Acad. Sci. USA* **118**, e2017525118 (2021).
- Terwilliger, T. C., Adams, P. D., Afonine, P. V. & Sobolev, O. V. A fully automatic method yielding initial models from high-resolution cryo-electron microscopy maps. *Nat. Methods* **15**, 905–908 (2018).
- Zhang, X., Zhang, B., Freddolino, P. L. & Zhang, Y. CR-I-TASSER: assemble protein structures from cryo-EM density maps using deep convolutional neural networks. *Nat. Methods* **19**, 195–204 (2022).
- Terashi, G., Wang, X., Prasad, D., Nakamura, T. & Kihara, D. DeepMainMast: integrated protocol of protein structure modeling for cryo-EM with deep learning and structure prediction. *Nat. Methods* **21**, 122–131 (2024).

9. Wang, X., Terashi, G. & Kihara, D. CryoREAD: de novo structure modeling for nucleic acids in cryo-EM maps using deep learning. *Nat. Methods* **20**, 1739–1747 (2023).
10. Liebschner, D. et al. Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. *Acta Crystallogr. D* **75**, 861–877 (2019).
11. Topf, M. et al. Protein structure fitting and refinement guided by cryo-EM density. *Structure* **16**, 295–307 (2008).
12. Rantos, V., Karius, K. & Kosinski, J. Integrative structural modeling of macromolecular complexes using Assemblin. *Nat. Protoc.* **17**, 152–176 (2022).
13. Lasker, K., Topf, M., Sali, A. & Wolfson, H. J. Inferential optimization for simultaneous fitting of multiple components into a CryoEM map of their assembly. *J. Mol. Biol.* **388**, 180–194 (2009).
14. Pettersen, E. F. et al. UCSF Chimera—a visualization system for exploratory research and analysis. *J. Comput. Chem.* **25**, 1605–1612 (2004).
15. Alnabati, E., Esquivel-Rodriguez, J., Terashi, G. & Kihara, D. MarkovFit: structure fitting for protein complexes in electron microscopy maps using Markov random field. *Front. Mol. Biosci.* **9**, 935411 (2022).
16. Han, X., Terashi, G., Christoffer, C., Chen, S. & Kihara, D. VESPER: global and local cryo-EM map alignment using local density vectors. *Nat. Commun.* **12**, 2090 (2021).
17. Berman, H. M. et al. The Protein Data Bank. *Nucleic Acids Res.* **28**, 235–242 (2000).
18. Varadi, M. et al. AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Res.* **50**, D439–D444 (2022).
19. Maddhuri Venkata Subramaniya, S. R., Terashi, G. & Kihara, D. Protein secondary structure detection in intermediate-resolution cryo-EM maps using deep learning. *Nat. Methods* **16**, 911–917 (2019).
20. Wang, X. et al. Detecting protein and DNA/RNA structures in cryo-EM maps of intermediate resolution using deep learning. *Nat. Commun.* **12**, 2302 (2021).
21. Mostosi, P., Schindelin, H., Kollmannsberger, P. & Thorn, A. Haruspex: a neural network for the automatic identification of oligonucleotides and protein secondary structure in cryo-electron microscopy maps. *Angew. Chem. Int. Ed.* **59**, 14788–14795 (2020).
22. Dhariwal, P. & Nichol, A. Diffusion models beat gans on image synthesis. *Adv. Neural Inf. Process. Syst.* **34**, 8780–8794 (2021).
23. Ho, J., Jain, A. & Abbeel, P. Denoising diffusion probabilistic models. *Adv. Neural Inf. Process. Syst.* **33**, 6840–6851 (2020).
24. Song, J., Meng, C. & Ermon, S. Denoising diffusion implicit models. In *Proc. International Conference on Learning Representations* (2021).
25. Ramesh, A., Dhariwal, P., Nichol, A., Chu, C. & Chen, M. Hierarchical text-conditional image generation with clip latents. Preprint at *arXiv* (2022).
26. Nichol, A. Q. et al. Glide: Towards photorealistic image generation and editing with text-guided diffusion models. In *Proc. 39th International Conference on Machine Learning* **162**, 16784–16804 (PMLR, 2023).
27. Wolleb, J., Sandkühler, R., Bieder, F., Valmaggia, P. & Cattin, P. C. Diffusion models for implicit image segmentation ensembles. In *Proc. 5th International Conference on Medical Imaging with Deep Learning* **172**, 1336–1348 (PMLR, 2022).
28. Chen, T., Li, L., Saxena, S., Hinton, G. & Fleet, D. J. A generalist framework for panoptic segmentation of images and videos. In *Proc. IEEE/CVF International Conference on Computer Vision* 909–919 (2023).
29. Saharia, C. et al. Palette: image-to-image diffusion models. In *ACM SIGGRAPH 2022 Conference Proceedings* 15 (Association for Computing Machinery, 2022).
30. Ruiz, N. et al. DreamBooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 22500–22510 (2023).
31. Corso, G., Jing, B., Barzilay, R. & Jaakkola, T. *International Conference on Learning Representations* (ICLR, 2023).
32. Watson, J. L. et al. De novo design of protein structure and function with RFdiffusion. *Nature* **620**, 1089–1100 (2023).
33. Yim, J. et al. SE (3) diffusion model with application to protein backbone generation. In *Proc. International Conference on Machine Learning* 1632 (JMLR, 2023).
34. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
35. Zhang, Y. & Skolnick, J. Scoring function for automated assessment of protein structure template quality. *Proteins* **57**, 702–710 (2004).
36. Sudre, C. H., Li, W., Vercauteren, T., Ourselin, S. & Jorge Cardoso, M. *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support* 240–248 (Springer, 2017).
37. Fontana, P. et al. Structure of cytoplasmic ring of nuclear pore complex by integrative cryo-EM and AlphaFold. *Science* **376**, eabm9326 (2022).
38. Dutta, D., Nguyen, V., Campbell, K. S., Padrón, R. & Craig, R. Cryo-EM structure of the human cardiac myosin filament. *Nature* **623**, 853–862 (2023).
39. Cramer, P. AlphaFold2 and the future of structural biology. *Nat. Struct. Mol. Biol.* **28**, 704–705 (2021).
40. Carreira-Perpinan, M. A. Acceleration strategies for Gaussian mean-shift image segmentation. In *Proc. IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'06)* 1160–1167 (IEEE, 2006).
41. Evans, R. et al. Protein complex prediction with AlphaFold-Multimer. Preprint at *bioRxiv* (2022).
42. Mukherjee, S. & Zhang, Y. MM-align: a quick algorithm for aligning multiple-chain protein complex structures using iterative dynamic programming. *Nucleic Acids Res.* **37**, e83 (2009).
43. He, J., Lin, P., Chen, J., Cao, H. & Huang, S.-Y. Model building of protein complexes from intermediate-resolution cryo-EM maps with deep learning-guided automatic assembly. *Nat. Commun.* **13**, 4066 (2022).
44. Afonine, P. V. et al. New tools for the analysis and validation of cryo-EM maps and atomic models. *Acta Crystallogr. D* **74**, 814–840 (2018).
45. Zhang, S., Li, N., Zeng, W., Gao, N. & Yang, M. Cryo-EM structures of the mammalian endo-lysosomal TRPML1 channel elucidate the combined regulation mechanism. *Protein Cell* **8**, 834–847 (2017).
46. Kucukelbir, A., Sigworth, F. J. & Tagare, H. D. Quantifying the local resolution of cryo-EM density maps. *Nat. Methods* **11**, 63–65 (2014).
47. Khan, A. K. et al. A steric ‘ball-and-chain’ mechanism for pH-mediated regulation of gap junction channels. *Cell Rep.* **31**, 107482 (2020).
48. Blees, A. et al. Structure of the human MHC-I peptide-loading complex. *Nature* **551**, 525–528 (2017).
49. Majumder, P. et al. Cryo-EM structures of the archaeal PAN-proteasome reveal an around-the-ring ATPase cycle. *Proc. Natl Acad. Sci. USA* **116**, 534–539 (2019).
50. Gutiérrez-Fernández, J. et al. Key role of quinone in the mechanism of respiratory complex I. *Nat. Commun.* **11**, 4135 (2020).
51. Ding, Z. et al. High-resolution cryo-EM structure of the proteasome in complex with ADP-AlFx. *Cell Res.* **27**, 373–385 (2017).
52. Ranson, N. A. et al. ATP-bound states of GroEL captured by cryo-electron microscopy. *Cell* **107**, 869–879 (2001).

53. Sah-Teli, S. K. et al. Structural basis for different membrane-binding properties of *E. coli* anaerobic and human mitochondrial β -oxidation trifunctional enzymes. *Structure* **31**, 812–825 (2023).
54. Sah-Teli, S. K. et al. Complementary substrate specificity and distinct quaternary assembly of the *Escherichia coli* aerobic and anaerobic β -oxidation trifunctional enzyme complexes. *Biochem. J.* **476**, 1975–1994 (2019).
55. Paul, D. M. et al. In situ cryo-electron tomography reveals filamentous actin within the microtubule lumen. *J. Cell Biol.* **219**, e201911154 (2020).
56. Webb, B. & Sali, A. Comparative protein structure modeling using MODELLER. *Curr. Protoc. Bioinformatics* **54**, 5.6.1–5.6.37 (2016).
57. Phillips, J. C. et al. Scalable molecular dynamics with NAMD. *J. Comput. Chem.* **26**, 1781–1802 (2005).
58. Jamali, K. et al. Automated model building and protein identification in cryo-EM maps. *Nature* <https://doi.org/10.1038/s41586-024-07215-4> (2024).
59. Altschul, S. F., Gish, W., Miller, W., Myers, E. W. & Lipman, D. J. Basic local alignment search tool. *J. Mol. Biol.* **215**, 403–410 (1990).
60. Takada, H. et al. RqcH and RqcP catalyze processive poly-alanine synthesis in a reconstituted ribosome-associated quality control system. *Nucleic Acids Res.* **49**, 8355–8369 (2021).
61. Turk, M. & Baumeister, W. The promise and the challenges of cryo-electron tomography. *FEBS Lett.* **594**, 3243–3261 (2020).
62. Chen, Z. et al. De novo protein identification in mammalian sperm using in situ cryoelectron tomography and AlphaFold2 docking. *Cell* **186**, 5041–5053.e5019 (2023).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2024

Methods

Constructing a benchmark dataset

Following the protocols employed in our previous works^{19,20,63,64}, we compiled a dataset of experimental cryo-EM maps for training, validation and testing DiffModeler. Initially, we sourced cryo-EM maps from EMDB (as of 26 January 2023) with resolutions between 5 Å and 10 Å and with the corresponding deposited structures in PDB with more than 20 residues. We only kept maps that contain only proteins. This initial screening yielded 840 maps.

Subsequently, we assessed the quality of structure-to-map fit by measuring cross-correlation and overlap between the EM maps and simulated maps generated from their respective structures in PDB¹⁷. Maps were discarded if their corresponding structures displayed a cross-correlation and overlap below 0.65. The remaining maps were manually inspected. These steps reduced the number of maps to 337.

To remove redundancy in the data, we applied single linkage clustering with the sequence identity of proteins within each map. Two maps were grouped into the same group if any protein chains from both maps exhibited a global sequence identity of 25% or higher. This clustering procedure resulted in 103 clusters. Out of the 103 clusters, we randomly allocated 68 clusters (230 maps) for the training set, 18 clusters (36 maps) for validation and 17 clusters (71 maps) for testing (Supplementary Table 1). It is important to note that the training, validation and testing sets are fully independent from each other. Finally, we classified maps in the testing set that contained inaccurate predicted models with a TM score lower than 0.5 in the AlphaFold Database¹⁸ and those that do not have such inaccurate models. Among the 71 maps, 19 maps did not have inaccurate models with a TM score lower than 0.5.

Preprocessing of map data

If a map had a grid size that is different from 1.0 Å, we interpolated the grid size to 1.0 Å using trilinear interpolation. The density values within a map were normalized to (0.0, 1.0) with a minimum–maximum normalization. Any negative values in a map were set to 0, and 0 was used as the minimum value for normalization. We set the maximum value for normalization as the 98th percentile density value, and any density values above that were capped at 1.0.

From each map, boxes of a size of 64³ Å³ were collected by scanning the box across a map along three axes with a stride of 32 Å. Each grid point within the box was assigned a label indicating whether it belonged to the backbone. If a grid point was within 2.0 Å of any backbone atoms, it was assigned as backbone. Otherwise, the point was considered as background. A box was excluded from training if less than 0.1% of the grid points were assigned as backbone.

Training the conditional diffusion model of DiffModeler

Given the density information from cryo-EM maps, the objective of the diffusion model of DiffModeler is to generate the backbone labels in the map. We employed a conditional diffusion model, the denoising diffusion implicit model²⁴, for its superior generation quality and efficiency. Inspired by the Pix2Seq²⁸ framework, we designed an encoder–decoder network architecture (Extended Data Fig. 1). The encoder scans the input density map with a box of 64³ size and embeds (outputs) hidden features of the map. The decoder utilizes three components as input of the conditional diffusion framework: the condition (the starting cryo-EM density map and hidden features), the noised backbone x_t at time step t and the time t of the current step. From these inputs, the decoder outputs the predicted traced backbone y_t . The noised backbone x_t is a mixture of the ground-truth traced backbone density x_0 and the Gaussian noise ϵ determined by the time step t , which will be explained later. The encoder and the decoder are optimized simultaneously by comparing the predicted traced backbone y_t and ground-truth traced backbone x_0 . The encoder and decoder neural

network architecture is shown in Extended Data Fig. 2a and 2b, respectively. The detailed network architecture of each component of the encoder and the decoder is shown in Extended Data Fig. 3.

As mentioned in the previous dataset section, we allocated 230 maps for training and 36 maps for validation of the conditional diffusion model. For each batch of training, we randomly sampled 8 boxes from the 230 maps. In total, around 16,000 and 3,500 boxes were used in an epoch for training and validation, respectively. The framework was trained through 30 epochs, and the final model is selected based on the validation performances.

The main objective of the model is to perform conditional denoising of a noisy density of the traced backbone to achieve the ground-truth traced protein backbone density x_0 in the map. For training the model, a series of noisy traced backbone density maps were generated by randomly sampling the density values from the ground-truth traced backbone density and the Gaussian noise

$$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon, \quad (1)$$

where

$$\alpha_t = \left(\cos \left(\frac{t + 0.0002}{1.00025} \times \frac{\pi}{2} \right) \right)^2, \quad (2)$$

where x_t is the noised traced backbone at time step t , α_t is a cosine scheduling function shown in equation (2), x_0 is the ground-truth traced backbone and ϵ is a noise variable randomly sampled from the standard Gaussian noise, $\mathcal{N}(\mathbf{0}, \mathbf{I})$. The ground-truth density of the traced protein backbone was prepared by assigning the backbone label to each grid point based on the corresponding backbone native structure (N, Ca and C atoms). For any grid point in the map, if a grid point was within 2.0 Å of any backbone atoms, it was assigned as backbone. Otherwise, the point was considered as background.

During the training process, t was uniformly sampled from (0, 1) for each map in the training set at each iteration to enforce that the framework successfully captures the diffusion process. The noised backbone x_t for time t was obtained according to equation (1), from which the decoder computes the predicted backbone map y_t . The loss of y_t was computed in comparison with the ground-truth backbone x_0 . The used Dice loss³⁶ was defined as

$$\begin{cases} L_{\text{Dice}} = 1 - \frac{2 \times \sum_{i=1}^N p_i g_i}{\sum_{i=1}^N p_i^2 + \sum_{i=1}^N g_i^2 + \epsilon} \\ L = \frac{1}{B} \sum_{k=1}^B L_{\text{Dice}}(k) \end{cases} \quad (3)$$

L_{Dice} represents the Dice loss of a predicted box P of prediction y_t at time step t and a corresponding ground-truth box G of ground truth x_0 ; N is the total number of grid points inside the box; $p_i \in P$ is the predicted probability of the i th grid point in the predicted box; $g_i \in G$ is the binary ground truth of the i th grid point, where 1 denotes the existence of backbone structure in the grid point and 0 indicates background; ϵ is a smoothing factor with a value of 1×10^{-6} ; L is the overall loss of a batch of B examples; and $L_{\text{Dice}}(k)$ represents the Dice loss of the k th example's detection. Here, different samples in the same batch may have different time step t since it is uniformly and independently sampled for each example.

We tested hyperparameter combinations of a learning rate of (1×10^{-3} , 1×10^{-4} , 1×10^{-5}) with a weight decay of (0 , 1×10^{-6} , 1×10^{-5} , 1×10^{-4}) using the Adam optimizer⁶⁵. Among the combinations, the learning rate 1×10^{-4} without weight decay showed the best grid-wise intersection-over-union of 0.562 on the validation set. Training and validation of the conditional diffusion model took around 5 days. The computations are performed on two paralleled NVIDIA RTX A6000 48 GB GPU connected via NVLink.

Inference of the conditional diffusion model in DiffModeler

With the trained conditional diffusion model, we compute the traced backbone conditioned on the input cryo-EM density. The inference of conditional diffusion model is shown in Extended Data Fig. 4. Given a box of cryo-EM density, the encoder of the conditional diffusion model first embeds the hidden features of the input density box. Subsequently, the decoder starts with the random Gaussian noise as the initial distribution x_T and iteratively refines the estimated density to make it closer to the ground-truth traced backbone x_0 , conditioned on the hidden features from the encoder and the initial density input.

Benefitting from the training, which used uniformly sampled timesteps, we have the flexibility to choose the overall inference steps T . We chose $T = 100$ as we did not observe significant performance improvement with T larger than 100. The current time step t is calculated by

$$t = 1 - \frac{i}{T}, \quad (4)$$

where t is the time step at inference iteration and T is the overall inference steps. Though the actual time step t as a fraction number belongs to $(0, 1)$ during inference, we simplify the term as integer in following description by starting $t = T$ to $t = 0$, which corresponds to the iteration $i = 0$ to $i = T$.

The first iteration of the inference starts at time step $t = T$. The decoder takes the random Gaussian noise x_T , time step T embedding and the condition (that is, the hidden feature embedding and the original cryo-EM map) as input and then it outputs y_T .

In the following time step $t = T - 1, T - 2, \dots, 0$, the condition inputs are the same and the time step t embedding obtained with equation (4). However, the noisy backbone input x_t for decoder is different from training. During training, x_t is computed following equation (1), which uses x_0 as the ground-truth traced backbone. As x_0 is not available in the inference stage, the input of the decoder, x_t , uses the decoder's output y_{t+1} at time step $t + 1$:

$$x_t = \sqrt{\alpha_t} \times y_{t+1} + \sqrt{1 - \alpha_t} \epsilon, \quad (5)$$

where x_t is the estimated noised backbone at time step t , y_{t+1} is the decoder output at $t + 1$ and ϵ is the random Gaussian noise. In this equation, ϵ is also estimated by comparing the decoder's noisy backbone input x_{t+1} and its corresponding backbone estimation output y_{t+1} from the decoder as follows:

$$\epsilon = \frac{1}{\sqrt{1 - \alpha_{t+1}}} \times (x_{t+1} - \sqrt{\alpha_{t+1}} \times y_{t+1}). \quad (6)$$

By combining equations (5) and (6), we can obtain the decoder input x_t with decoder output y_{t+1} at time step $t = T - 1, T - 2, \dots, 0$. The inference process of the decoder is repeated for $T = 100$ times and x_0 at time step $t = 0$ is our final estimated backbone.

Single-chain structure fitting using VESPER

We used VESPER¹⁶ for fitting AF2 models of individual proteins to the modified map by the diffusion model. AF2 models of the protein chains were taken from the AlphaFold database¹⁸. Supplementary Table 3 provides the TM scores of the chains. The average TM score was 0.922. The fitting process involved three main steps: initially, AF2 models were transformed into simulated maps at a 1 Å resolution using TEMPY⁶⁶. In the subsequent step, we simplified both the modified EM map and the simulated maps of the AF2 models into maps by condensing them into maps with local representative density points. This was achieved through the mean-shifting algorithm⁴⁰, a method we devised in our early work, MAINMAST⁴. Finally, VESPER was used to globally align AF2 models into various poses within the representative map, generating

different fit scores. The top 100 poses were retained as pose candidates for each subunit.

The mean-shift algorithm is employed to compute maps featuring local representative density points by clustering density points within an EM map. First, grid points with a density exceeding 0 are identified. Then, the algorithm iteratively updates the coordinates of a grid point x by considering the weights associated with neighboring grid points $x^{t+1} = f(x^t)$, where

$$f(x) = \frac{\sum_{x_i \in N(x)} K(x - x_i) \phi(x_i) x_i}{\sum_{x_i \in N(x)} K(x - x_i) \phi(x_i)}. \quad (7)$$

$N(x)$ is the neighborhood of x , which are a set of neighboring grid points that satisfy $\|x_i - x\|_2 \leq 2 \times \sigma$; $K(p)$ is a Gaussian kernel function with bandwidth σ , as shown in equation (8); $\phi(x)$ is the density value of the grid point x .

$$K(p) = \exp\left(-1.5 \left\| \frac{p}{\sigma} \right\|^2\right), \quad (8)$$

where p is the distance between two points, the σ is the bandwidth set as 2. The mean-shift process is continued until convergence, that is, $\|x^{t+1} - x^t\|_2 \leq \delta$ with δ set to 0.001.

Following the completion of the mean-shifting process, we merged shifted points that were in close proximity. Points closer than a pre-defined threshold distance of 2.0 Å, were clustered together and the grid point with the highest density within the cluster was designated as the representative node. This clustering and selection process was iterated until the convergence of the selected representative nodes. The resulting set of points, known as representative points, forms the basis for the representative map (Fig. 3).

By completing this stage, we acquired two distinct representative maps using the mean-shift algorithm: the subunit representative map (RM_{subunit}) derived from the simulated map of the AF2 single-chain structure and the backbone representative map (RM_{backbone}) obtained from the diffusion-traced backbone map.

The final step involves utilizing VESPER to globally align AF2 single-chain subunits into various poses within the backbone map. Specifically, VESPER aligns different RM_{subunit} to RM_{backbone} obtained in the preceding step. For each subunit representative map RM_{subunit} , VESPER systematically explores all potential poses to align RM_{subunit} with RM_{backbone} . In VESPER's global search, we used a rotation scan interval of 10° and a translation scan interval of 2 Å. The fitness score of RM_{subunit} i at pose j is defined as

$$\text{subunit_fitscore}(i, j) = \frac{P}{N}, \quad (9)$$

where P is the number of Cα positions of subunit at pose j that have representative points in RM_{backbone} within 3 Å and N is the total number of Cα positions of subunit. The top 100 poses were kept for each subunit. This pool comprises $M \times 100$ pose candidates for a protein structure complex with M chains.

Assembling subunits to generate the entire protein complex structure

Subunits, fitted to the map with different pose candidates, are then assembled into a complete protein complex structure model. We developed a greedy algorithm that iteratively assembles superimposed subunits within the map. The entire pipeline is depicted in Extended Data Fig. 5. As outlined in the preceding section, we generated 100 poses for each subunit in the map using VESPER. Therefore, the subunit–pose pool for a given protein structure complex comprises $M \times 100$ pose candidates, all of which were scored using the subunit_fitscore (equation (9)).

Table 1 | The statistical information of Fig. 2

Figure	Method	Regression line	Pearson correlation coefficient	P value	S.e.m.
Fig. 2d	DiffModeler	$y = -0.012x + 0.999$	-0.110	0.655	0.027
Fig. 2d	EMBuild	$y = -0.095x + 1.449$	-0.360	0.131	0.060
Fig. 2d	VESPER (raw)	$y = -0.225x + 1.850$	-0.716	0.001	0.053
Fig. 2d	Phenix	$y = -0.097x + 1.031$	-0.379	0.110	0.058
Fig. 2e	DiffModeler	$y = 9.22 \times 10^{-6}x + 0.872$	0.434	0.063	4.64×10^{-6}
Fig. 2e	EMBuild	$y = 2.69 \times 10^{-6}x + 0.827$	0.053	0.829	1.23×10^{-5}
Fig. 2e	VESPER (raw)	$y = -3.68 \times 10^{-5}x + 0.605$	-0.608	0.006	1.16×10^{-5}
Fig. 2e	Phenix	$y = -1.65 \times 10^{-5}x + 0.498$	-0.336	0.160	1.13×10^{-5}
Fig. 2f	DiffModeler	$y = 1.48 \times 10^{-5}x + 0.815$	0.433	0.064	7.45×10^{-6}
Fig. 2f	EMBuild	$y = -1.63 \times 10^{-5}x + 0.830$	-0.246	0.310	1.56×10^{-5}
Fig. 2f	VESPER (raw)	$y = -4.23 \times 10^{-5}x + 0.537$	-0.603	0.006	1.36×10^{-5}
Fig. 2f	Phenix	$y = -2.71 \times 10^{-5}x + 0.433$	-0.448	0.054	1.31×10^{-5}
Fig. 2g	DiffModeler	$y = 3.59 \times 10^{-5}x + 3.700$	0.135	0.581	6.38×10^{-5}
Fig. 2g	EMBuild	$y = 7.65 \times 10^{-5}x + 3.672$	0.146	0.551	1.26×10^{-3}
Fig. 2g	VESPER (raw)	$y = 9.93 \times 10^{-4}x + 4.743$	0.755	1.87×10^{-4}	2.09×10^{-4}
Fig. 2g	Phenix	$y = 9.54 \times 10^{-4}x + 5.348$	0.790	5.67×10^{-5}	1.79×10^{-4}

The initial step in the modeling process involves selecting the subunit–pose with the highest subunit_fit_score among all available poses. Subsequently, a local region within 20 Å from the fitted subunit–pose is masked out in the backbone map $RM_{backbone}$ and the subunit–pose is further optimized in terms of the subunit_fit_score with an interval of 5° for rotation scan and an interval of 1 Å for translation scan in that local region. Then, from the subunit–pose pool, subunit–poses are removed if the poses belong to the subunit that was just selected or if they have large overlap with the selected subunit–pose. A subunit–pose is considered to have overlap if more than 10% of Cα positions of the subunit–pose are closer than 3 Å to any Cα positions to an already selected subunit–pose(s).

Following this, the subsequent best subunit–pose is selected iteratively until the subunit–pose pool is exhausted. In most cases, where each subunit assumes a correct pose, all M subunits are successfully fitted into the map. However, there are rare instances where not all subunits are selected due to large overlap among all 100 poses of a subunit with other already selected subunit–poses. In such scenarios, where some subunits remain unfitted due to substantial overlap, a new pose set is generated for these remaining subunits. This is achieved by fitting them to the remaining density regions within $RM_{backbone}$ using VESPER. The same iterative process is then applied until all the subunits are successfully fitted.

In the output cif file of the structure model, subunit_fit_score is shown in the occupancy field of each residue of chains.

Domain-based DiffModeler

The results obtained from DiffModeler fitting are inevitably influenced by the (in)accuracy of AF2 models of proteins included in the maps. The impact of inaccuracies of AF2 models can often be mitigated by fitting individual domains of proteins because AF2 tends to build accurate domain structures even when relative orientations of domains are incorrect. To address this, we have implemented a procedure to cut multiple chain structure models into individual domains using SWORD2 (ref. 67) and use the domains in the fitting process. Extended Data Fig. 10 shows an example where domain-based fitting improved

the model accuracy. The domain-based DiffModeler is also available at our GitHub codebase and server. Alternatively, AF2 models could be also trimmed to remove low-confidence regions, which is an area left for future work.

Evaluation metrics

Backbone recall. The backbone recall was computed for each residue by determining the fraction of backbone heavy atoms within a 3 Å proximity to any grid points in the diffused map. This was then averaged across all residues in the map.

Backbone precision. The backbone precision was computed as the fraction of grid points within a 3 Å proximity to any backbone atoms.

To evaluate the performance of modeled structure, we utilized MM-align⁴² to compare the modeled structure and the native structure. MM-align is a sequence-independent alignment of protein complex structures, which aims to find the best superposition between two protein complex structures via a heuristic iteration of a modified Needleman–Wunsch dynamic programming algorithm. The heuristic alignment procedure is repeated until the alignment between two protein complexes converges.

Given a modeled protein complex with M residues and a native protein complex with N residues, and the number of aligned residues is K identified by MM-align, then the evaluation metrics are calculated as follows:

TM score. The structural similarity between modeled structure and native structure.

$$TM - Score = \frac{1}{N} \sum_{i=1}^K \frac{1}{1 + d_{ij}^2/d_0^2(N)}, \quad (10)$$

where N is the total number of residues in the native structure and K is the number of aligned residues. d_{ij} is the distance between the Cα atoms of the residue and its aligned pair j from the modeled structure after superposition by MM-align and $d_0(L) = 1.24\sqrt{N-15} - 1.8$ specified by MM-align.

Align ratio. The fraction of residues in the native structure that have aligned residues from the modeled structure.

$$Align_ratio = \frac{K}{N}, \quad (11)$$

where N is the total number of residues in the native structure and K is the number of aligned residues identified by the alignment algorithm MM-align.

Sequence identity. The fraction of residues that are in the native structure that have aligned residues from the modeled structure and the aligned residues have the same residue type.

$$SeqID = \frac{L}{N}, \quad (12)$$

where L is the number of residues that have correct residue types among all K aligned residues identified by MM-align and N is the total number of residues in the native structure.

The r.m.s.d. The r.m.s.d. between K aligned residues of the modeled structure and native structure.

$$r.m.s.d. = \sqrt{\frac{1}{K} \sum_{i=1}^K \delta_i^2}, \quad (13)$$

where K is the number of aligned residues identified by MM-align and δ_i is the Euclidean distance between the C α atoms of the aligned residues.

Statistical information of the benchmark dataset

In the following table, we present the statistical information of the benchmark dataset shown in Fig. 2 (Table 1).

Software used for the benchmark dataset

The software used in the benchmark dataset is Phenix-v1.21.1-5286, VESPER-vpub1, EMBuild-v1.0 and DiffModeler-v1.0.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The entries of the maps and corresponding structure models utilized in this study are provided in Supplementary Tables 1, 4 and 6. The experimental EM maps utilized can be downloaded from the EMDB (<https://www.emdataresource.org/>). The corresponding experimental determined structures utilized can be downloaded from the Research Collaboratory for Structural Bioinformatics (RCSB) PDB (<https://www.rcsb.org/>). The structures modeled by DiffModeler, diffused maps, intermediate diffusion results and the corresponding native structures from RCSB are available via Zenodo at <https://zenodo.org/records/12155184> (ref. 68). The single-chain AF2-predicted structures are from the AlphaFold Database (<https://alphafold.ebi.ac.uk/>). Source data are provided with this paper.

Code availability

The source code of DiffModeler is made available via GitHub at <https://github.com/kiharalab/DiffModeler> (ref. 69). It can run on our web-server <https://em.kiharalab.org/algorithm/DiffModeler> freely without installing it in a local machine. We also provide sequence version of DiffModeler on our server [https://em.kiharalab.org/algorithm/DiffModeler\(seq\)](https://em.kiharalab.org/algorithm/DiffModeler(seq)), which can automatically use the sequence information to find the most similar single-chain structure from RCSB and AlphaFold database and then model the full protein complex structure. The source code of ComplexModeler (including DiffModeler and CryoREAD) for protein–DNA/RNA complex structure modeling is made available via GitHub at <https://github.com/kiharalab/ComplexModeler>. It is also available via our webserver at <https://em.kiharalab.org/algorithm/ComplexModeler>. All the code is also available via Zenodo at <https://doi.org/10.5281/zenodo.13132116> (ref. 68).

References

63. Terashi, G., Wang, X., Prasad, D., Nakamura, T. & Kihara, D. DeepMainMast: integrated protocol of protein structure modeling for cryo-EM with deep learning and structure prediction. *Nat. Methods* **21**, 122–131 (2023).
64. Wang, X., Terashi, G. & Kihara, D. De novo structure modeling for nucleic acids in cryo-EM maps using deep learning. *Nat. Methods* **20**, 1739–1747 (2023).
65. Kingma, D. P. & Ba, J. Adam: A method for stochastic optimization. In *Proc. 3rd International Conference on Learning Representations* (eds Bengio, Y. & LeCun, Y.) (2015).
66. Farabella, I. et al. TEMPy: a Python library for assessment of three-dimensional electron microscopy density fits. *J. Appl. Crystallogr.* **48**, 1314–1323 (2015).
67. Cretin, G. et al. SWORD2: hierarchical analysis of protein 3D structures. *Nucleic Acids Res.* **50**, W732–W738 (2022).
68. Wang, X., Zhu, H., Terashi, G., Taluja, M. & Kihara, D. Data of ‘DiffModeler: large macromolecular structure modeling for cryo-EM maps using diffusion model’. Zenodo <https://doi.org/10.5281/zenodo.12155184> (2024).
69. Wang, X., Zhu, H., Terashi, G., Taluja, M. & Kihara, D. Code of ‘DiffModeler: large macromolecular structure modeling for cryo-EM maps using diffusion model’. Zenodo <https://doi.org/10.5281/zenodo.13132116> (2024).

Acknowledgements

We thank J. C. Verburgt, A. Jain and C. Christoffer for their help in literature search, discussion and proofreading. We also thank J. A. Nash, S. Ellis and J. Chen for their suggestion for optimizing the released software. This work was partly supported by the National Institutes of Health (R01GM133840) and the National Science Foundation (DMS2151678, DBI2003635, CMM1825941, MCB2146026 and MCB1925643). X.W. is recipient of the MolSSI graduate fellowship.

Author contributions

D.K. conceived the study. X.W. designed and implemented DiffModeler and computed the results. H.Z. and G.T. optimized the VESPER algorithm and participated in implementing the full pipeline. All the authors analyzed the results. X.W. drafted the manuscript and D.K. edited it. All the authors read and approved the manuscript.

Competing interests

Authors declare that they have no competing interests.

Additional information

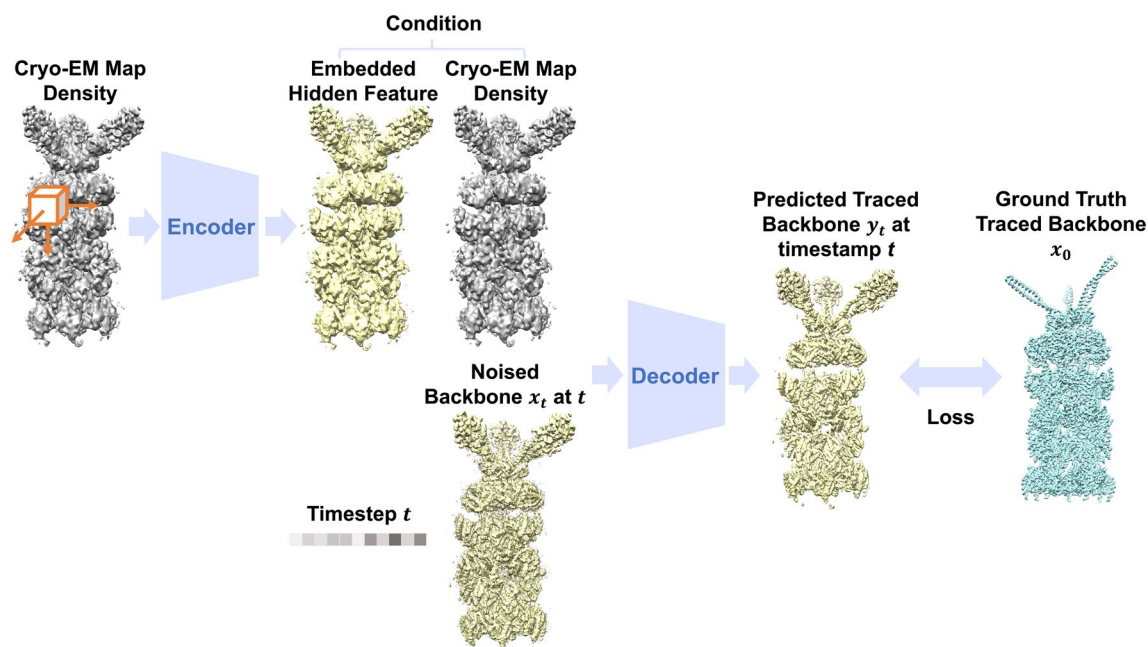
Extended data is available for this paper at <https://doi.org/10.1038/s41592-024-02479-0>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41592-024-02479-0>.

Correspondence and requests for materials should be addressed to Daisuke Kihara.

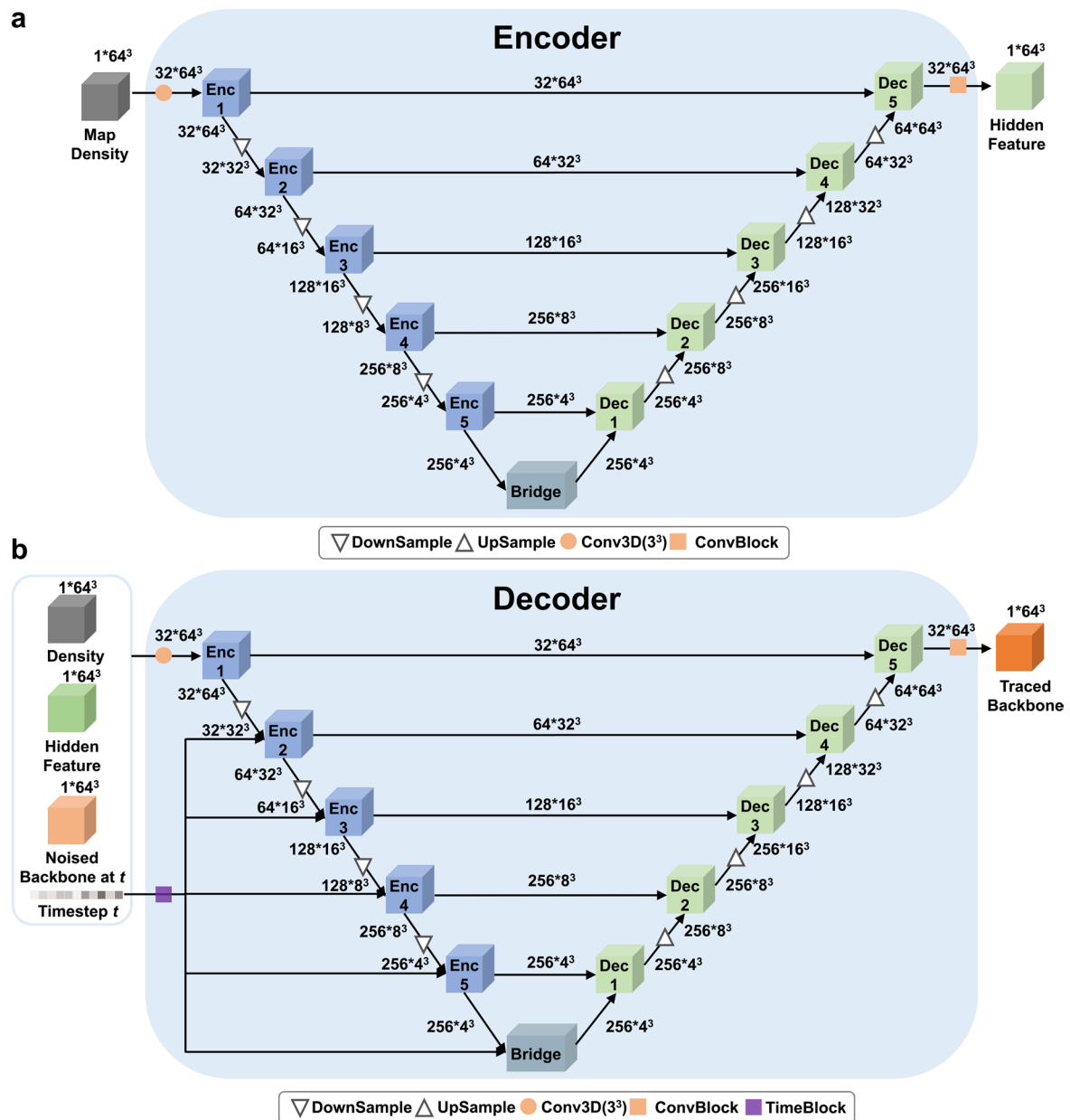
Peer review information *Nature Methods* thanks Matthew Belousoff and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Arunima Singh, in collaboration with the *Nature Methods* team. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.



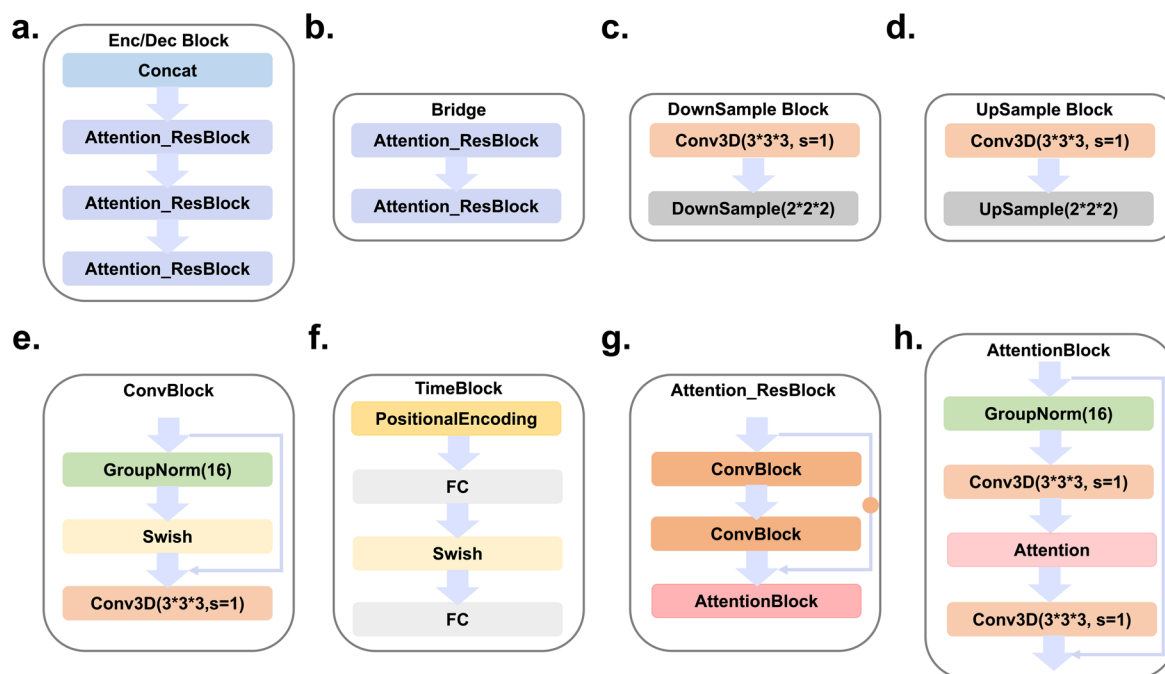
Extended Data Fig. 1 | The overall framework of the conditional diffusion model in DiffModeler. The entire framework consists of one encoder and one decoder. The encoder takes the cryo-EM density as input and outputs the hidden features by scanning the map density with a box. The decoder utilizes three main components as the input of the conditional diffusion framework: the condition

(the starting cryo-EM density map and hidden features), the noised backbone x_t at timestep t , and the timestep t . Then the decoder outputs the predicted traced backbone y_t . The encoder and the decoder are optimized simultaneously by comparing the predicted traced backbone y_t and ground truth traced backbone x_0 , with details illustrated in Methods.



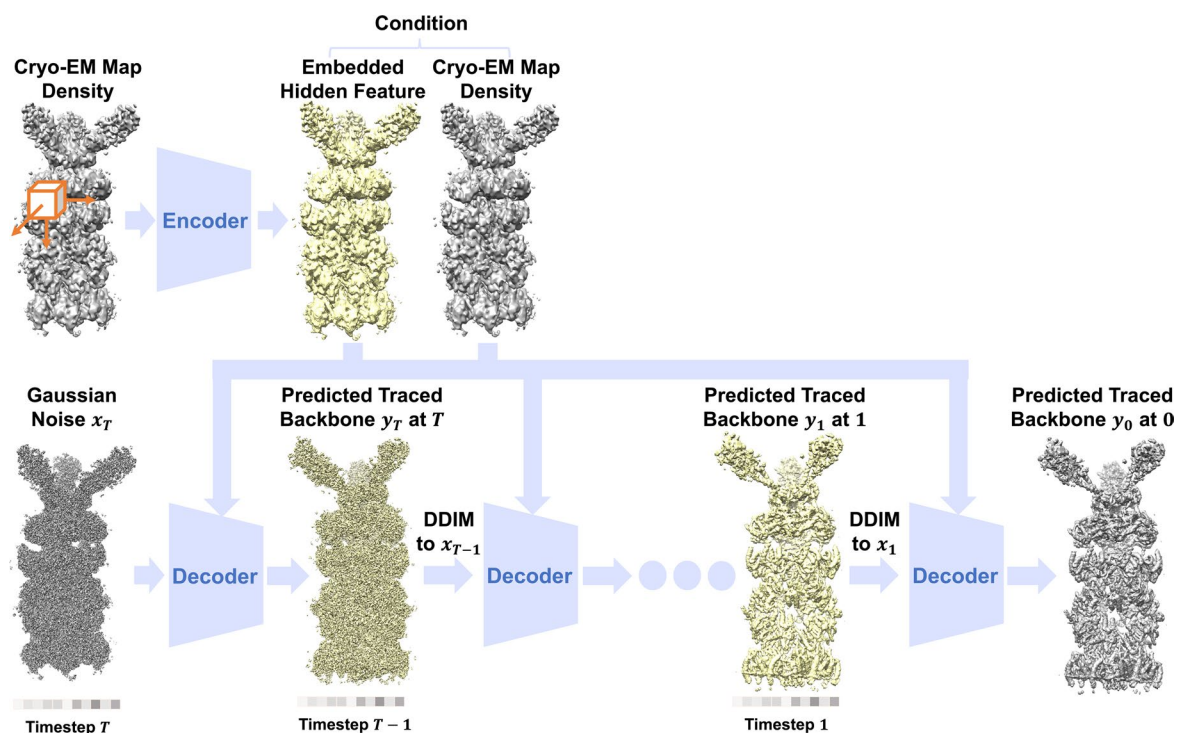
Extended Data Fig. 2 | The network architecture of the conditional diffusion model in DiffModeler. a. The encoder network architecture. It is a 3D U-shape-based convolutional neural network (UNet) with skip connections. The channel size of different layers is also illustrated in the figure. The input is first processed by Conv3D layer with 32 filters in size of 3^3 , and then iteratively processed and down-sampled by encoding block Enc1-Enc5 (Extended Data Fig. 3a), The downsample block (Extended Data Fig. 3c), the dense information is further processed by bridge block (Extended Data Fig. 3b), subsequently process the encoding, which is upsampled by Dec1-Dec5 (Extended Data Fig. 3a). The

upsample blocks (Extended Data Fig. 3d) with skip connections connecting with the The network architecture of the conditional diffusion model in DiffModeler encoding blocks, and the final ConvBlock (Extended Data Fig. 3e) aggregate the information and yield the final output. **b.** The decoder network architecture. It shares a similar UNet architecture as the encoder. Additionally, it includes a TimeBlock (Extended Data Fig. 3f) that encodes the timestep input and passes it to every levels of encoding block in the decoder network. Individual blocks are illustrated in Extended Data Fig. 3.



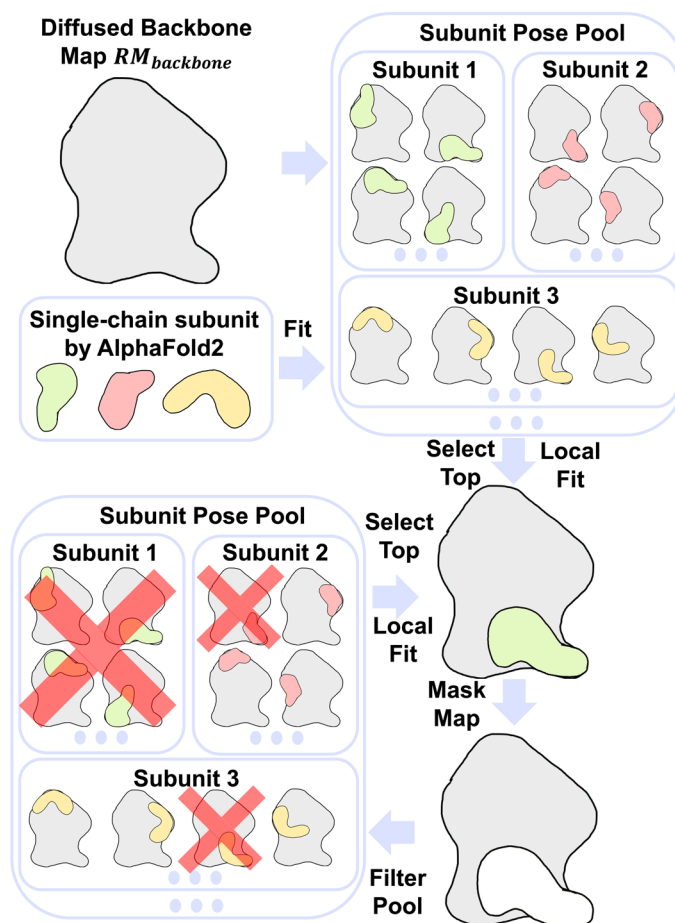
Extended Data Fig. 3 | Individual network block architecture of the conditional diffusion model in DiffModeler. **a.** The encoder/decoder block (Enc1-Enc5, Dec1-Dec5, in panel a and b of Extended Data Fig. 2). Concat is an operation that concatenates inputs. **b.** The bridge block (located at the bottom of Extended Data Fig. 2a,b); **c.** The DownSample Block. Conv3D is a 3-dimensional (3D) convolutional layer with a filter size of $3 \times 3 \times 3$, stride 1, and padding 1. **d.** The UpSample Block. **e.** The ConvBlock (located one step before the output box in Extended Data Fig. 2a, S2b). GroupNorm is a normalization layer that calculates group statistics across channels to normalize the input data by dividing multiple channels into different groups. Swish is a smooth, non-monotonic function that

consistently matches or outperforms ReLU and serves as an activation layer. **f.** Time Block, specifically designed for timestep embedding. PositionalEncoding is an explicit layer with pairs of sine and cosine functions to add positional information to the input. FC is a fully connected layer in which each neuron applies a linear transformation to the input vector through a weight matrix. **g.** Attention_ResBlock (Attention_ResBlock in panel a, b); **h.** The Attention Block (AttentionBlock in panel g). Attention is a layer that enables to dynamically highlight the relevant features of the input data through the attention mechanism.



Extended Data Fig. 4 | The inference pipeline of the conditional diffusion model. During the inference stage, the encoder first takes the cryo-EM density as input and outputs the hidden features. Then, the decoder iteratively refines the density from time step T utilizing three core information as input to estimate the traced backbone: condition (cryo-EM density and hidden features), the noised

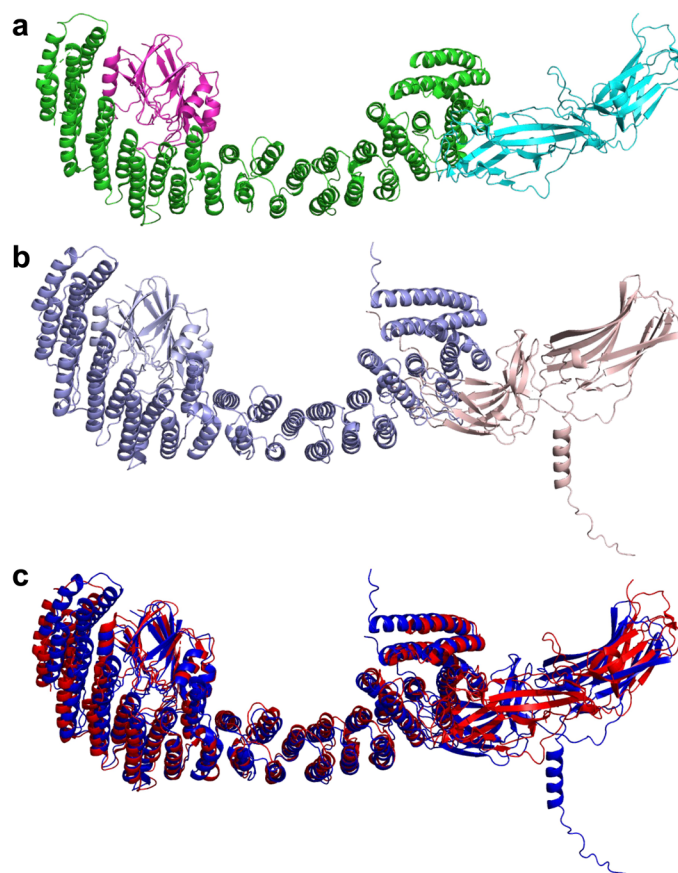
traced backbone at timestep t , the embedding of timestep t . The noised traced backbone starts with a random Gaussian noise x_T and x_t at timestep t is iteratively updated by the decoder's output y_{t+1} through DDIM step (illustrated in Methods).



Extended Data Fig. 5 | The overall pipeline of assembling algorithm.

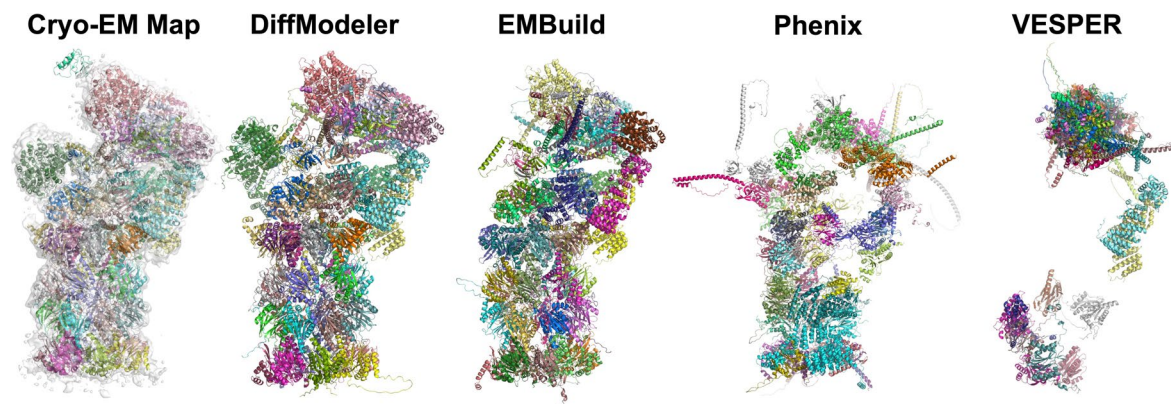
Simulated maps of AlphaFold2 models for each subunit are aligned with the $RM_{backbone}$ using VESPER, and the top 100 poses for each subunit are cataloged in the structure pose pool. Initially, the subunit-pose exhibiting the highest *subunit_fit_score* is chosen from the structure pool. The local density region within the map occupied by this subunit is then masked out (the white local region in the figure).

The pose of the selected subunit undergoes further refinement by VESPER, employing smaller angle and shifting intervals. Subsequent subunit-poses in the pool are eliminated under two conditions as shown by red crosses in the figure: if they belong to the same subunit as the one just selected or if they overlap with the selected subunit-pose. This iterative process continues until all the subunits are chosen to construct the full complex, at which point the pool becomes empty.



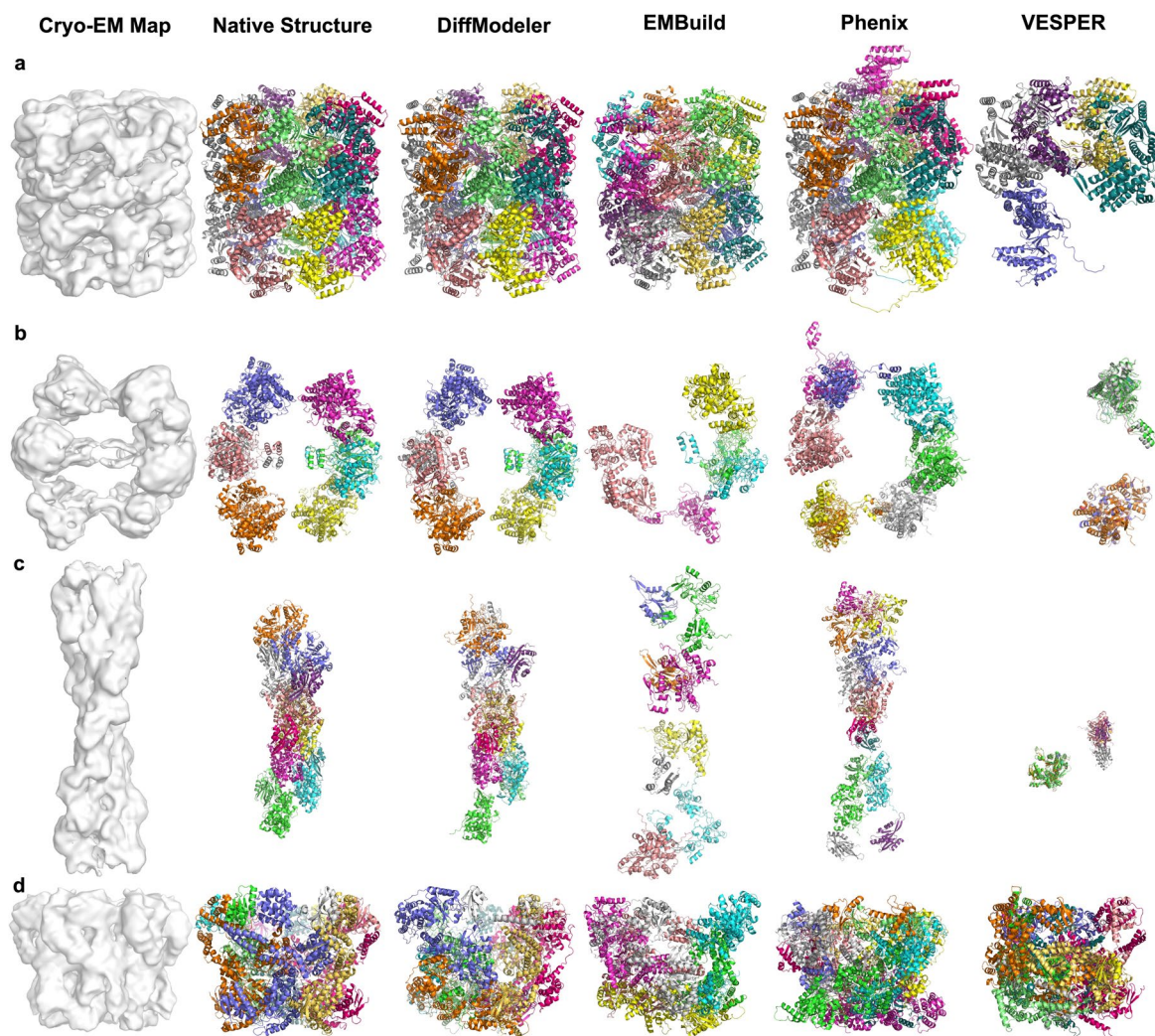
Extended Data Fig. 6 | Example of estimated fitting quality for modeled structure. **a.** native structure of Mouse retromer (VPS26/VPS35/VPS29) heterotrimer (EMD-21136, PDB ID: 6VAC, Resolution: 5.70 Å; protein lengths: 3 chains and 1,202 amino acids (aa)). Different colors represent different chains. **b.** modeled structures colored by subunit_fit_score, scaled from blue to red for

high to low scores. **c.** the superposition of the model by DiffModeler (blue) with the native structure (red): TM-Score: 0.66, Sequence Identity: 0.54, RMSD: 4.96 Å. The chain colored in red on the right in the panel b do not have a correct pose relative to the native structure.



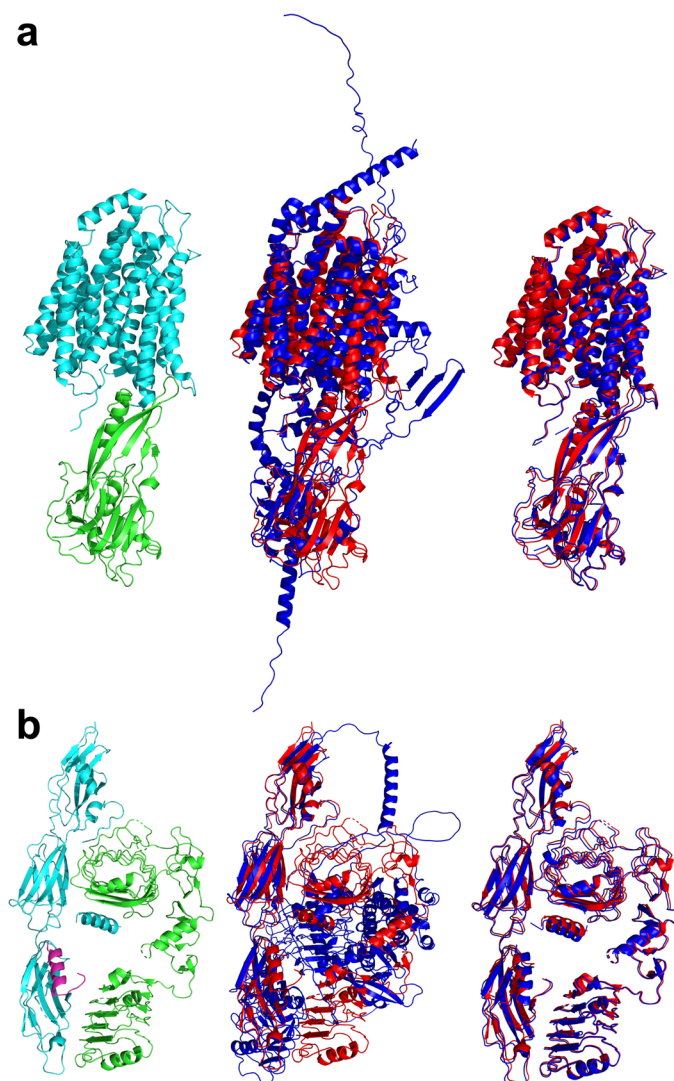
Extended Data Fig. 7 | Atomic structure modeling by different methods for experimental map EMD-6872. The proteasome in complex with ADP-AIFx (EMD-6693, PDB ID: 5WV1, Resolution: 6.30 Å; protein lengths: 47 chains and 13,462 amino acids (aa)). The 5 columns from left to right are 1) EM map and its corresponding structure; 2) the atomic structure by DiffModeler: TM-Score:

0.94, Align Ratio: 0.96, Sequence Identity: 0.89, RMSD: 5.13 Å; 3) the atomic structure by EMBuild: TM-Score: 0.88, Align Ratio: 0.91, Sequence Identity: 0.47, RMSD: 4.8 Å. 4) the atomic structure by Phenix: TM-Score: 0.38, Align Ratio: 0.51, Sequence Identity: 0.04, RMSD: 17.4 Å; 5) the atomic structure by VESPER: TM-Score: 0.25, Align Ratio: 0.29, Sequence Identity: 0.16, RMSD: 12.5 Å.



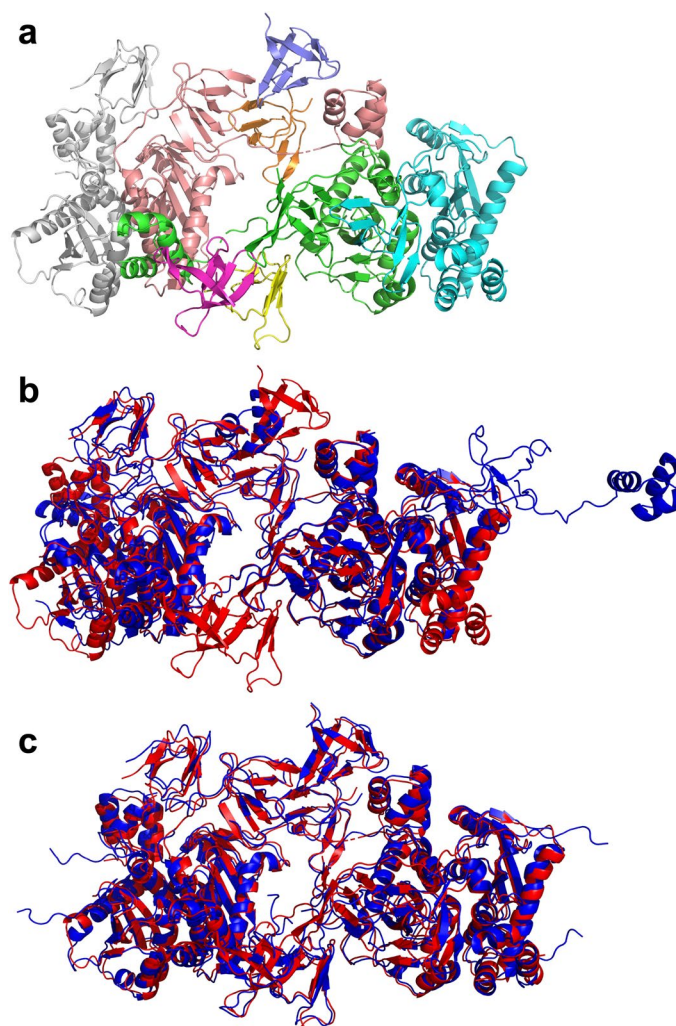
Extended Data Fig. 8 | Examples of structure models built with three other methods for experimental maps at low resolution (10–15 Å). Detailed Evaluation Results are shown in Sup Table S4. In each row of the modeling example, five columns shown from left to right are 1) input cryo-EM map; 2) the corresponding native structure; 3) the structure model by DiffModeler; 4) the structure model by EMBuild; 5) the structure model by Phenix; 6) the structure model by VESPER (raw). The DiffModeler model and its superposition is also shown in Fig. 5. **a.** ATP-Bound States of GroEL (EMD-1042, PDB ID: 1GR5, Resolution: 10.3 Å; protein lengths: 14 chains and 7,238 amino acids (aa)): DiffModeler (EMBuild, Phenix, VESPER): TM-Score: 0.97 (0.96, 0.78, 0.37), Sequence Identity: 0.95 (0.92, 0.71, 0.34), RMSD: 3.88 Å (4.90 Å, 6.50 Å, 7.76 Å). **b.** acid beta oxidation trifunctional enzyme (anEcTFE) octameric complex

(EMD-16134, PDB ID: 8BNR, Resolution: 10.3 Å; protein lengths: 8 chains and 4,584 aa): DiffModeler (EMBuild, Phenix, VESPER): TM-Score: 0.87 (0.30, 0.30, 0.19), Sequence Identity: 0.88 (0.23, 0.17, 0.16), RMSD: 5.80 Å (9.65 Å, 11.28 Å, 8.03 Å). **c.** cofilactin filament inside microtubule lumen (EMD-16877, PDB ID: 8OH4, Resolution: 16.5 Å; protein lengths: 14 chains and 3,776 aa): DiffModeler (EMBuild, Phenix, VESPER): TM-Score: 0.60 (0.17, 0.25, 0.18), Sequence Identity: 0.50 (0.02, 0.05, 0.08), RMSD: 8.30 Å (12.1 Å, 13.19 Å, 12.67 Å). **d.** MecA-ClpC complex with ATP with the Walker B mutations introduced in the D2 ring (EMD-5608, PDB ID: 3J3S, Resolution: 11.0 Å; protein lengths: 12 chains and 5,352 aa): DiffModeler (EMBuild, Phenix, VESPER): TM-Score: 0.51 (0.26, 0.26, 0.48), Sequence Identity: 0.45 (0.16, 0.15, 0.41), RMSD: 8.42 Å (12.28 Å, 11.91 Å, 8.72 Å).



Extended Data Fig. 9 | Examples of Modeled Structure by DiffModeler (AF2) and DiffModeler (native). From left to right, we present the native structure, with different colors represent different chains, superposition of native structure (red) and modeled structure (blue) by DiffModeler using AF2 single-chain structures, superposition of native structure (red) and modeled structure (blue) by DiffModeler using native single-chain structures **a.** thermostable human MFSD2A in complex with thermostable human Sync2 (EMD-12935, PDB ID: 7OIX,

Resolution: 3.60 Å; protein lengths: 2 chains and 716 aa); DiffModeler (AF2): TM-Score: 0.63, Sequence Identity: 0.57, RMSD: 5.01 Å; DiffModeler (native): TM-Score: 0.99, Sequence Identity: 1.00, RMSD: 0.74 Å. **b.** insulin receptor (IR) bound with S597 component 2 (EMD-27705, PDB ID: 8DTM, Resolution: 3.50 Å; protein lengths: 3 chains and 802 aa); DiffModeler (AF2): TM-Score: 0.54, Sequence Identity: 0.54, RMSD: 4.09 Å; DiffModeler (native): TM-Score: 0.99, Sequence Identity: 1.00, RMSD: 0.78 Å.

**Extended Data Fig. 10 | Example of Modeled Structure by fitting domains.**

a. the native protein structure the core MMTV intasome (EMD-6441, PDB ID: 3JCA, Resolution: 4.80 Å; protein lengths: 8 chains and 1,226 aa). Different colors represent different chains. This entry is included in the CryoREAD dataset.

b. the superposition of the model by original DiffModeler (blue) with the native structure (red): TM-Score: 0.65, sequence identity: 0.64, RMSD: 4.37 Å **c.** the superposition of the model by domain-based DiffModeler (blue) with the native structure (red): TM-Score: 0.87, sequence identity: 0.85, RMSD: 2.05 Å.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- | | | |
|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☒ | ☐ | The statistical test(s) used AND whether they are one- or two-sided
<i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i> |
| ☒ | ☐ | A description of all covariates tested |
| ☒ | ☐ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☒ | ☐ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

We did not use any software for data collection. The data is directly downloaded from EMDB and RCSB.

Data analysis

The program is freely available for academic use via Github, <https://github.com/kiharalab/DiffModeler> and our server: <https://em.kiharalab.org/algorithm/DiffModeler>. The software used in benchmark is Phenix-v1.21.1-5286, VESPER-vpub1, EMBuild-v1.0, and DiffModeler-v1.0.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The entries of the maps and corresponding structure models are provided in Supplementary Table 1-6. The experimental EM maps can be downloaded from EMDB (<https://www.emdataresource.org/>). The corresponding experimental determined structures can be downloaded from RCSB (<https://www.rcsb.org/>). Structure

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	N/A
Reporting on race, ethnicity, or other socially relevant groupings	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The data we used in this work is from two public databases, PDB and EMD. The procedure of selecting entries for the datasets we used are described in Methods in details at the section titled "Constructing benchmark dataset" and the subsequent sections. First, we collected all the map-structure pairs that are in the specified resolution range. Then we analyzed the overlap, correlation of the map-structure pairs to filter out reliable map-structure pair for training/testing. Finally we utilized the sequence identity to make sure the independence of the testing set relative to training set to benchmark our methods.
Data exclusions	Data exclusion are based on correlation, sequence identity and so on with details in "Constructing benchmark dataset" sections in methods
Replication	User can easily use our public open source code and our online server to reproduce all results in this paper.
Randomization	It is not relevant to our research since our samples are not allocated into experimental groups.
Blinding	We randomly dividing data for training and testing, with details discussed in "Constructing benchmark dataset" section. We also have two additional independent datasets for independent testing. During this process, we were blinded to group allocation during data collection and/or analysis.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

N/A

Novel plant genotypes

N/A

Authentication

N/A